

Automorphisms of reductive groups

SGA 3, Exposé XXIV

English draft

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About this draft

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Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XXIV. Automorphisms of reductive groups

⁽⁰⁾The first part of this exposé (nos. 1 to 5) is a direct consequence of the existence, for a reductive group, of “sufficiently many outer automorphisms”, a result which is a consequence of the weakest form of the isomorphism theorem for pinned groups. The second part (nos. 6 and 7) presents two applications of the more precise results of the preceding exposé; in particular, no. 7 uses the theorem on generators and relations in its explicit form. Finally, in an appendix (no. 8) we have given results on “Galois” cohomology used in the text.

Let us specify our cohomological notation: if S is a scheme and G an S -group scheme, we shall denote by $H^1(S, G)$ the first cohomology set of S with coefficients in G , computed for the (fpqc) topology; it is also the set of isomorphism classes of (fpqc) principal homogeneous sheaves under G . We shall denote by $H_{\text{ét}}^1(S, G)$ the corresponding set for the étale topology; it is therefore the part of $H^1(S, G)$ consisting of the classes of homogeneous sheaves under G which are quasi-isotrivial (= locally trivial for the étale topology). We shall denote by $\text{Fib}(S, G)$ the part of $H^1(S, G)$ consisting of the classes of representable sheaves (principal homogeneous bundles). We therefore have the inclusions

$$H_{\text{ét}}^1(S, G) \subset H^1(S, G), \quad \text{Fib}(S, G) \subset H^1(S, G).$$

If every principal homogeneous sheaf under G is representable (for example if G is quasi-affine over S , cf. SGA 1, VIII 7.9), we therefore have $\text{Fib}(S, G) = H^1(S, G)$.

If $S' \rightarrow S$ is a covering morphism for the (fpqc) topology, we denote by $H^1(S'/S, G)$ the kernel of the canonical map $H^1(S, G) \rightarrow H^1(S', G_{S'})$. One knows that $H^1(S'/S, G)$ can be computed simplicially (TDTE I, § A.4), which implies that when $S' \rightarrow S$ is covering for the étale topology, $H^1(S'/S, G)$ is also the kernel of $H_{\text{ét}}^1(S, G) \rightarrow H_{\text{ét}}^1(S', G_{S'})$.

Finally, following Exp. VIII, 4.5, one calls “theorem 90” the following assertion: “every principal homogeneous sheaf under $\mathbf{G}_{m,S}$ is representable and locally trivial”, an assertion equivalent to “ $H^1(S, \mathbf{G}_{m,S}) = \text{Pic}(S)$ ”, or again to “ $H^1(S, \mathbf{G}_{m,S}) = 0$ for S local (or, more generally, semi-local)”.

1. Scheme of automorphisms of a reductive group

1.0. — It is appropriate first to specify certain definitions from the preceding exposé. Let S be a nonempty scheme, G a reductive S -group, and $\mathcal{R} = (M, M^*, R, R^*, \Delta)$ a reduced pinned root datum (Exp. XXIII 1.5). One calls a *pinning of G of type $\mathcal{R}$* , or an *$\mathcal{R}$ -pinning of G* , the datum:

- (i) of an isomorphism of $D_S(M)$ onto a maximal torus T of G (or, equivalently, of a monomorphism $D_S(M) \rightarrow G$ whose image is a maximal torus T of G), identifying R with a root system of G relative to T (Exp. XIX, 3.6) and R^* with the set of corresponding coroots,
- (ii) for each $\alpha \in \Delta$, of an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

For G to possess an $\mathcal{R}$ -pinning, it is necessary and sufficient that it be splittable and of type $\mathcal{R}$ (Exp. XXII, 2.7).

If $u : G \rightarrow G'$ is an isomorphism of reductive S -groups, to every $\mathcal{R}$ -pinning $\mathcal{E}$ of G there corresponds, by “transport of structure”, an $\mathcal{R}$ -pinning $u(\mathcal{E})$ of G' . If $v : \mathcal{R}' \rightarrow \mathcal{R}$ is an

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

isomorphism of pinned root data, to every $\mathcal{R}'$ -pinning $\mathcal{E}$ of G' there corresponds by transport of structure an $\mathcal{R}$ -pinning $v(\mathcal{E})$ of G .

Let us call a *pinned group* a triple $(G, \mathcal{R}, \mathcal{E})$ where G is a reductive S -group, $\mathcal{R}$ a reduced pinned root datum, and $\mathcal{E}$ a pinning of G of type $\mathcal{R}$. One calls an *isomorphism* of the pinned group $(G, \mathcal{R}, \mathcal{E})$ onto the pinned group $(G', \mathcal{R}', \mathcal{E}')$ a pair (u, v) where u is an isomorphism $u : G \rightarrow G'$ and v an isomorphism of pinned root data $v : \mathcal{R}' \rightarrow \mathcal{R}$, such that $u(\mathcal{E}) = v(\mathcal{E}')$.⁽¹⁾

N. B. If S is nonempty, v is uniquely determined by u , and by abuse of language one will also say that u is an isomorphism of the pinned groups. In particular, if $(G, \mathcal{R}, \mathcal{E})$ is a pinned group, an automorphism of $(G, \mathcal{R}, \mathcal{E})$ is therefore an automorphism u of G such that there exists an automorphism v of $\mathcal{R}$ such that $u(\mathcal{E}) = v(\mathcal{E})$; it is therefore an automorphism of G , normalizing T , inducing on T an automorphism of the form $D_S(h)$, where h is an automorphism of M , and permuting the elements X_α , $\alpha \in \Delta$, among themselves. (As one readily sees, the preceding conditions in fact characterize the automorphisms of $(G, \mathcal{R}, \mathcal{E})$.)

One has an evident contravariant functor

$$\Phi : (G, \mathcal{R}, \mathcal{E}) \mapsto \mathcal{R}, \quad (u, v) \mapsto v$$

and the principal result of the preceding exposé (Exp. XXIII, 4.1) shows us that it is a fully faithful functor (we shall moreover see in the following exposé that it is an equivalence of categories). It follows in particular that the group of automorphisms of $(G, \mathcal{R}, \mathcal{E})$ is canonically isomorphic to the group of automorphisms of the pinned root datum $\mathcal{R}$ (cf. Exp. XXIII, 5.5).

1.1. — Let S be a scheme; equip $(\text{Sch})/S$ with the (fpqc) topology and consider the S -sheaf of groups $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, where G is an S -group scheme. One has an exact sequence of S -sheaves of groups

$$1 \longrightarrow \underline{\text{Centr}}(G) \longrightarrow G \xrightarrow{\text{int}} \underline{\text{Aut}}_{S\text{-gr.}}(G)$$

which defines a monomorphism

$$j : G/\underline{\text{Centr}}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G).$$

The image sheaf of j is the sheaf of inner automorphisms of G ; for an automorphism u of G to be inner, it is necessary and sufficient that there exist a covering family $\{S_i \rightarrow S\}$ and, for each i , a $g_i \in G(S_i)$ such that $\text{int}(g_i) = u_{S_i}$. In this case, if v is another automorphism of G , one sees at once that $\text{int}(v)u = vu v^{-1}$ is the inner automorphism defined by the family $g'_i = v(g_i)$. It follows that the image of j is normal in $\underline{\text{Aut}}_{S\text{-gr.}}(G)$. The quotient sheaf of groups, denoted by $\underline{\text{Autext}}(G)$, is the sheaf of outer automorphisms of G . One therefore has an exact sequence

$$1 \longrightarrow G/\underline{\text{Centr}}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\text{Autext}}(G) \longrightarrow 1.$$

The preceding definitions are all compatible with base change. They are of course valid in any site.

1.2. — Let S be a scheme and $(G, \mathcal{R}, \mathcal{E})$ a pinned reductive S -group. Let E be the (abstract) group of automorphisms of the pinned root datum $\mathcal{R}$, i.e. the group of automorphisms of $\mathcal{R}$ normalizing Δ . By Exp. XXIII, 5.5, one has a canonical monomorphism

$$E \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G)$$

which associates to $h \in E$ the unique automorphism u of the pinned group G such that $\Phi(u) = h$. This monomorphism canonically defines a monomorphism of sheaves

$$(*) \quad a : E_S \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G).$$

For an automorphism u of G to be a section of the image sheaf of a , it is necessary and sufficient that the following conditions hold:

⁽¹⁾**N.D.E.** Thus $\text{Lie}(u)(X_{v(\alpha)}) = X'_\alpha$, for every $\alpha \in R'$.

- (i) u normalizes T . One then knows that u permutes the roots of G relative to T . If $\alpha \in R$, then $u(\alpha) : t \mapsto \alpha(u^{-1}(t))$ is therefore locally on S of the form $t \mapsto \beta(t)$, with $\beta \in R$. The second condition can then be written as follows:
- (ii) If $\alpha \in \Delta$ and if U is an open subset of S such that $u(\alpha)_U \in R$, then $u(\alpha)_U \in \Delta$ and

$$\mathrm{Lie}(u_U)(X_\alpha)_U = (X_{u(\alpha)})_U.$$

It follows at once from the definitions that the sections of $a(E_S)$ normalize the subgroups of G defined by the pinning: T, B, B^-, U, U^- . These definitions being given, one has:

Theorem 1.3. *Let S be a scheme and G a reductive S -group. Consider the canonical exact sequence of S -sheaves of groups⁽²⁾*

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \xrightarrow{p} \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

- (i) $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable by a smooth separated S -scheme.
- (ii) $\underline{\mathrm{Autext}}(G)$ is representable by a twisted constant S -scheme of finite generation (Exp. X, 5.1).
- (iii) If G is splittable, the preceding exact sequence is split. More precisely, for every pinning of G , the morphism (cf. 1.2 (*))

$$p \circ a : E_S \longrightarrow \underline{\mathrm{Autext}}(G)$$

is an isomorphism.

Let us first show how the theorem follows from the following lemma:

Lemma 1.4. *Under the hypotheses of (iii), $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is the semidirect product $a(E_S) \cdot \mathrm{ad}(G)$.⁽³⁾*

The lemma immediately implies the theorem when G is splittable. Since G is locally splittable for the étale topology (Exp. XXII, 2.3), hence also for the (fppf) topology, and since the latter is “of effective descent” for the fibered category of twisted constant morphisms (Exp. X, 5.5), one deduces (ii) in the general case (cf. Exp. IV, 4.6.8). To deduce (i), one observes that $\mathrm{ad}(G)$ is affine over S , hence the morphism p is affine when $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable, and one concludes by descent of affine schemes.⁽⁴⁾

It therefore only remains for us to prove 1.4. To do so, it suffices to prove:

Lemma 1.5. *If $(\mathcal{R}, \mathcal{E})$ and $(\mathcal{R}', \mathcal{E}')$ are two pinning of the reductive S -group G , there exists a unique inner automorphism u of G over S taking one pinning to the other (i.e. such that there exists $v : \mathcal{R}' \xrightarrow{\sim} \mathcal{R}$ such that $u(\mathcal{E}) = v(\mathcal{E}')$, cf. 1.0).*

1.5.1. — Uniqueness. It suffices to prove that if G is a pinned S -group and if $\mathrm{int}(g)$ is an automorphism of a pinned group ($g \in G(S)$), then $\mathrm{int}(g) = \mathrm{id}$. Now one first has $\mathrm{int}(g)T = T$, $\mathrm{int}(g)B = B$, hence $g \in \underline{\mathrm{Norm}}_G(T)(S) \cap \underline{\mathrm{Norm}}_G(B)(S) = T(S)$ (cf. for example Exp. XXII, 5.6.1). It follows that $\mathrm{int}(g)$ normalizes each U_α and that

$$\mathrm{Lie}(\mathrm{int}(g))X_\alpha = \mathrm{Ad}(g)X_\alpha = \alpha(g)X_\alpha$$

for every $\alpha \in \Delta$. One therefore has $\alpha(g) = 1$ for $\alpha \in \Delta$, hence $g \in \bigcap_{\alpha \in \Delta} (\mathrm{Ker} \alpha)(S) = \underline{\mathrm{Centr}}(G)(S)$ (Exp. XXII, 4.1.8). Q.E.D.

1.5.2. — Existence. It suffices to prove it locally for the (fpqc) topology. Let

$$(T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta}) \quad \text{and} \quad (T', M', R', \Delta', (X'_{\alpha'})_{\alpha' \in \Delta'})$$

⁽²⁾**N.D.E.** Recall that $\mathrm{ad}(G) = G/\underline{\mathrm{Centr}}(G)$ denotes the adjoint group of G , cf. XXII 4.3.6.

⁽³⁾**N.D.E.** Here and in what follows we have replaced the notation $\mathrm{int}(G)$ by $\mathrm{ad}(G)$.

⁽⁴⁾**N.D.E.** Cf. SGA 1, VIII 2.1.

be the two pinnings. By conjugacy of maximal tori, one may suppose $T = T'$. After restricting S , one may suppose that the isomorphism $D_S(M) \simeq D_S(M')$ comes from an isomorphism $M \simeq M'$ taking R onto R' , and one is reduced to the situation $T = T'$, $M = M'$, $R = R'$. Since the systems of simple roots are conjugate under the Weyl group (Exp. XXI, 3.3.7), one may also suppose $\Delta = \Delta'$. There then exists for each $\alpha \in \Delta$ a scalar $z_\alpha \in \mathbf{G}_m(S)$ such that $X'_\alpha = z_\alpha X_\alpha$, and it suffices to construct locally for (fpqc) a section t of T such that $\alpha(t) = z_\alpha$ for each $\alpha \in \Delta$. But the morphism $T \rightarrow (\mathbf{G}_{m,S})^\Delta$ with components $\{\alpha, \alpha \in \Delta\}$ is the dual of an *injection* $\mathbf{Z}^\Delta \rightarrow M$, hence is faithfully flat, which completes the proof of 1.5.2 and therefore of 1.3.

Corollary 1.6. *Let S be a scheme and G a reductive S -group. The following conditions are equivalent:*

- (i) $\underline{\text{Aut}}_{S\text{-gr.}}(G)$ is affine (resp. of finite type, resp. of finite presentation, resp. quasi-compact) over S .
- (ii) $\underline{\text{Autext}}(G)$ is finite over S .
- (iii) For every $s \in S$, one has $\text{rkred}(G_s) - \text{rkss}(G_s) \leq 1$.

PROOF. Indeed, since $\text{ad}(G)$ is affine, flat and of finite presentation over S , the morphism $p : \underline{\text{Aut}}_{S\text{-gr.}}(G) \rightarrow \underline{\text{Autext}}(G)$ is affine, faithfully flat and of finite presentation.

If $\underline{\text{Autext}}(G)$ is finite over S , it is affine over S , hence so is $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, which proves (ii) $\Rightarrow$ (i). If $\underline{\text{Aut}}_{S\text{-gr.}}(G)$ is quasi-compact over S , it is of finite presentation over S (being in any case locally of finite presentation and separated over S); by Exp. V, 9.1, $\underline{\text{Autext}}(G)$ is then of finite presentation over S , hence finite, which proves (i) $\Rightarrow$ (ii). Finally, to prove the equivalence of (ii) and (iii), one may suppose G split, and one is reduced to Exp. XXI, 6.7.8. $\square$

Corollary 1.7. *Let S be a scheme and G a reductive S -group. Then one has $\underline{\text{Aut}}_{S\text{-gr.}}(G)^0 \simeq \text{ad}(G)$.*

Corollary 1.8. *Let S be a scheme, G a reductive S -group, and H a smooth affine S -group scheme with connected fibers over S . Then the S -functor*

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, H)$$

is representable by a smooth separated S -scheme (which is affine over S if G is semisimple).

Indeed, let U be the set of points s of S such that H_s is reductive; it is open (Exp. XIX, 2.6); if S' is an S -scheme, $H_{S'}$ is reductive if and only if $S' \rightarrow S$ factors through U . It follows that the canonical morphism $\underline{\text{Isom}}_{S\text{-gr.}}(G, H) \rightarrow S$ factors through U . One may therefore suppose $S = U$ and one is reduced to:

Corollary 1.9. *Let S be a scheme and G, G' two reductive S -groups. Then*

$$F = \underline{\text{Isom}}_{S\text{-gr.}}(G, G')$$

is representable by a smooth separated S -scheme (affine if G or G' is semisimple). Moreover, S decomposes as the sum of two open subschemes S_1 and S_2 such that $F_{S_1} = \emptyset$ and F_{S_2} is a left (resp. right) principal homogeneous bundle under $\underline{\text{Aut}}_{S_2\text{-gr.}}(G'_{S_2})$ (resp. $\underline{\text{Aut}}_{S_2\text{-gr.}}(G_{S_2})$).

PROOF. Indeed, let S_2 be the set of points of S where G and G' have the same type, and let S_1 be its complement.

Since the type of a reductive group is a locally constant function, S_1 and S_2 are open. It is clear that $F_{S_1} = \emptyset$, and one may suppose $S = S_2$. By Exp. XXIII, 5.6, F is a principal homogeneous sheaf under $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, locally trivial for the étale topology. It follows that $F_0 = F/\text{ad}(G)$ is a principal homogeneous sheaf under $\underline{\text{Autext}}(G)$, locally trivial for the étale topology, hence representable (Exp. X, 5.5). Since $\text{ad}(G)$ is affine, F is therefore also representable. $\square$

Let us observe that in the course of the proof one has obtained:

Corollary 1.10. *Let S be a scheme and G, G' two reductive S -groups of the same type at every $s \in S$. Then $\mathrm{ad}(G)$ acts freely (on the right) on $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$, and the quotient sheaf*

$$\underline{\mathrm{Isomext}}(G, G') = \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') / \mathrm{ad}(G)$$

is representable by a twisted constant S -scheme, which is a principal homogeneous bundle under $\underline{\mathrm{Autext}}(G)$ (and which is therefore finite over S if G is semisimple). Moreover, the isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \simeq \underline{\mathrm{Isom}}_{S\text{-gr.}}(G', G)$$

defined by $u \mapsto u^{-1}$ induces an isomorphism

$$\underline{\mathrm{Isomext}}(G, G') \simeq \underline{\mathrm{Isomext}}(G', G).$$

Remark 1.11. If $\underline{\mathrm{Isomext}}(G, G')(S) \neq \emptyset$, one says that G is an inner twisted form of G' ; then G' is an inner twisted form of G ; one can then reduce the structure group of $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ to $\mathrm{ad}(G)$. More precisely, let $u \in \underline{\mathrm{Isomext}}(G, G')(S)$, considered as a section $u : S \rightarrow \underline{\mathrm{Isomext}}(G, G')$. Denote by

$$\underline{\mathrm{Isomint}}_u(G, G')$$

the inverse image, under the canonical morphism $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \rightarrow \underline{\mathrm{Isomext}}(G, G')$, of the closed subscheme of $\underline{\mathrm{Isomext}}(G, G')$ which is the image of u . The natural action of $\mathrm{ad}(G)$ on $\underline{\mathrm{Isomint}}_u(G, G')$ equips this scheme with a structure of principal homogeneous bundle; by extension of the structure group⁽⁵⁾ $\mathrm{ad}(G) \rightarrow \underline{\mathrm{Aut}}(G)$, $\underline{\mathrm{Isomint}}_u(G, G')$ gives back $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$.

By Hensel's lemma (Exp. XI, 1.11), 1.8 immediately gives:

Corollary 1.12. *Let S be a henselian local scheme, G a reductive S -group, G' a smooth affine S -group with connected fibers, and s the closed point of S . If G_s and G'_s are isomorphic algebraic $\kappa(s)$ -groups, G and G' are isomorphic. More precisely, every $\kappa(s)$ -isomorphism $G_s \simeq G'_s$ comes from an S -isomorphism $G \simeq G'$.*

Applying 1.7 (resp. 1.12) to the scheme of dual numbers over a field, one deduces from Exp. III, 2.10 (resp. 3.10) part (i) (resp. (ii)) of the following corollary.

Corollary 1.13. *Let k be a field and G a reductive k -group.*

- (i) *If G is adjoint, one has $H^1(G, \mathrm{Lie}(G/k)) = 0$.⁽⁶⁾*
- (ii) *One has $H^2(G, \mathrm{Lie}(G/k)) = 0$.*

Remark 1.14. (i) The assertion concerning H^1 was known (Chevalley); that concerning H^2 was proved by Chevalley in most cases of the classification.

(ii) In fact, the conjunction of 1.13 and the uniqueness theorem over an algebraically closed field is essentially equivalent to the uniqueness theorem. A direct proof of 1.13 would therefore give a way to deduce the general uniqueness theorem from Chevalley's uniqueness theorem over a field.

The existence of reductive groups of all types over all schemes (Exp. XXV) shows that the obstructions to lifting a reductive k -group G over Artinian rings with residue field k (which, by Exp. III, 3.8, are elements of

$$H^3(G, \mathrm{Lie}(G/k) \otimes V) \simeq H^3(G, \mathrm{Lie}(G/k)) \otimes V,$$

⁽⁵⁾**N.D.E.** We have corrected $\mathrm{ad}(G) \rightarrow \underline{\mathrm{Autext}}(G)$ to $\mathrm{ad}(G) \rightarrow \underline{\mathrm{Aut}}(G)$.

⁽⁶⁾**N.D.E.** We have corrected the original, which stated (i) without assuming G adjoint. According to the characterization of the $H^i(G, -)$ as derived functors of $\mathrm{Hom}_G(k, -)$ (Exp. I, 5.3.1, see also [Ja87], I 4.16), one has $H^i(G, V) = \mathrm{Ext}_G^i(k, V)$ for every G -module V ; now if $\mathrm{char}(k) = p > 0$ and if $G = \mathrm{SL}_{p,k}$, then $\mathrm{Lie}(\mathrm{GL}_p/k)$ is a nontrivial extension of k by $\mathfrak{g} = \mathrm{Lie}(\mathrm{SL}_p/k)$, hence $H^1(G, \mathfrak{g}) \neq 0$. See also the addition 1.15.1 below.

where V is a certain k -vector space (equipped with the trivial action of G)) vanish. This seems to suggest that $H^3(G, \text{Lie}(G/k)) = 0$. Here again, a direct proof of this fact (if it is true) would doubtless give a way to deduce the general existence theorem from the existence theorem over a field (Chevalley's Tôhoku).

Corollary 1.15. *Let k be a field,⁽⁷⁾ and G a reductive k -group. Consider k as a trivial G -module. Then*

$$H^1(G, k) = H^2(G, k) = 0.$$

PROOF. ⁽⁸⁾Since $H^i(G_{\bar{k}}, \bar{k}) = H^i(G, k) \otimes \bar{k}$, one may suppose k algebraically closed. An element of $H^1(G, k)$ is nothing other than a morphism of k -groups $\phi : G \rightarrow \mathbf{G}_{a,k}$. Then $\phi(G) = G/\text{Ker}(\phi)$ is a smooth, connected and reductive subgroup (cf. XIX 1.7) of $\mathbf{G}_{a,k}$, hence trivial. Thus $H^1(G, k) = 0$.

Consider now the reductive k -group $H = G \times_k \mathbf{G}_{m,k}$. One has $\text{Lie}(H/k) = \text{Lie}(G/k) \oplus k$, a decomposition stable under H . For an arbitrary H -module V , one has

$$H^i(H, V) = H^i(G, H^0(\mathbf{G}_{m,k}, V))$$

(this follows from the characterization of the $H^i(H, -)$ as derived functors of $H^0(H, -) = H^0(G, H^0(\mathbf{G}_{m,k}, -))$, and from the fact that the functor $H^0(\mathbf{G}_{m,k}, -)$ is exact, cf. Exp. I, 5.3.1 and 5.3.3). In particular, for every i one has

$$H^i(H, \text{Lie}(H/k)) = H^i(G, \text{Lie}(G/k)) \oplus H^i(G, k)$$

hence $H^2(G, k) = 0$ by applying 1.13 (ii) to the reductive group H . $\square$

Corollary 1.15.1. ⁽⁹⁾*Let k be a field, G a reductive k -group, Z its center and $\mathcal{R} = (M, M^*, R, R^*)$ the type of G . Then one has*

$$H^1(G, \text{Lie}(G/k)) \simeq \text{Ext}_{\mathbf{Z}}^1(M/\Gamma_0(R), k).$$

In particular, $H^1(G, \text{Lie}(G/k)) = 0$ if and only if Z is smooth over k .

PROOF. Indeed, let $\mathfrak{g}$ (resp. $\mathfrak{z}$, resp. $\mathfrak{g}_{\text{ad}}$) be the Lie algebra of G (resp. Z , resp. $G_{\text{ad}} = G/Z$). It follows from 1.15 (and its proof) that

$$H^1(G, \mathfrak{g}) = H^1(G_{\text{ad}}, \mathfrak{g}) \simeq H^1(G_{\text{ad}}, \mathfrak{g}/\mathfrak{z}).$$

Put $C = \text{Coker}(\mathfrak{g} \rightarrow \mathfrak{g}_{\text{ad}})$; one has an exact sequence

$$0 \longrightarrow \mathfrak{g}/\mathfrak{z} \longrightarrow \mathfrak{g}_{\text{ad}} \longrightarrow C \longrightarrow 0.$$

Since $H^1(G_{\text{ad}}, \mathfrak{g}_{\text{ad}}) = 0$ (1.13) and $H^0(G_{\text{ad}}, \mathfrak{g}_{\text{ad}}) = 0$ (cf. Exp. II, 5.2.3), one obtains

$$H^1(G_{\text{ad}}, \mathfrak{g}/\mathfrak{z}) \simeq H^0(G_{\text{ad}}, C).$$

To compute the right-hand term, one may suppose k algebraically closed. Let (G, T, M, R) be a splitting of G ; then $Z = D_k(M/\Gamma_0(R))$ and C identifies with

$$\text{Coker}(\text{Hom}_{\mathbf{Z}}(M, k) \rightarrow \text{Hom}_{\mathbf{Z}}(\Gamma_0(R), k)) \simeq \text{Ext}_{\mathbf{Z}}^1(M/\Gamma_0(R), k),$$

equipped with the trivial action of G_{ad} . The corollary follows, since Z is not smooth over k if and only if $\text{char}(k) = p > 0$ and $M/\Gamma_0(R)$ has p -torsion (Exp. IX, 2.1). $\square$

⁽⁷⁾**N.D.E.** We have replaced “scheme” by “field”.

⁽⁸⁾**N.D.E.** Because of the correction made in 1.13, we have given another proof in the case of H^1 . Moreover, it follows from a theorem of G. Kempf that $H^i(G, k) = 0$ for every $i > 0$, cf. [Ja87], II 4.5 and 4.11.

⁽⁹⁾**N.D.E.** We have added this corollary, drawn from remarks of O. Gabber, which makes 1.13 (i) more precise.

Definition 1.16. Let S be a scheme and G a reductive S -group. One calls a *form of G over S* an S -group scheme G' locally isomorphic to G for the (fpqc) topology (it amounts to the same thing to say (cf. Exp. XXIII, 5.6) that G' is locally isomorphic to G for the étale topology, or again that G' is a reductive S -group of the same type as G at every point of S).

Corollary 1.17. *Let S be a scheme and G a reductive S -group.*

(i) *The functor*

$$G' \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$$

is an equivalence between the category of forms of G over S and the category of principal homogeneous bundles under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$.

(ii) *If $S' \rightarrow S$ is a covering morphism, forms of G trivialized by S' and bundles trivialized by S' correspond to one another.*

(iii) *Every principal homogeneous sheaf under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable and quasi-isotrivial (i.e. locally trivial for the étale topology).*

PROOF. The first assertion is formal in the category of sheaves (for (fpqc), for example). Moreover, every sheaf locally isomorphic to G (for (fpqc)) is representable (since G is affine over S) and locally isomorphic to G for the étale topology. Finally, for every form G' of G , the S -sheaf $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ is representable, by 1.8. The corollary follows at once. $\square$

Corollary 1.18. *The set of isomorphism classes of forms of the reductive group G over S is isomorphic to*

$$H^1(S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)) = H^1_{\text{ét}}(S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)) = \mathrm{Fib}(S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)).$$

If $S' \rightarrow S$ is a covering morphism, the subset consisting of forms trivialized by S' is isomorphic to $H^1(S'/S, \underline{\mathrm{Aut}}_{S\text{-gr.}}(G))$.

Corollary 1.19. *Let S be a scheme and $\mathcal{R}$ a reduced pinned root datum such that $\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})^{(10)}$ exists (a condition automatically satisfied, cf. Exp. XXV). Put*

$$\begin{aligned} A_S(\mathcal{R}) &= \underline{\mathrm{Aut}}_{S\text{-gr.}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})) \\ &= \mathrm{ad}(\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})) \cdot E(\mathcal{R})_S. \end{aligned}$$

(i) *The set of isomorphism classes of reductive S -groups of type $\mathcal{R}$ (Exp. XXII, 2.7) is isomorphic (by Exp. XXIII, 5.12) to*

$$H^1(S, A_S(\mathcal{R})) = H^1_{\text{ét}}(S, A_S(\mathcal{R})) = \mathrm{Fib}(S, A_S(\mathcal{R})).$$

(ii) *If $S' \rightarrow S$ is a covering morphism, the subset consisting of classes of groups splittable over S' is isomorphic to $H^1(S'/S, A_S(\mathcal{R}))$.*

Remark 1.20. With the preceding notation, to every reductive S -group of type $\mathcal{R}$ there is canonically associated a right principal homogeneous bundle under $A_S(\mathcal{R})$:

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R}), G) = P.$$

Observe that P can be interpreted as the “scheme of pinnings of G of type $\mathcal{R}$ ” (cf. 1.0). Moreover P is also a principal homogeneous bundle (on the left) under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$, a structure which appears immediately in the above description.

Proposition 1.21. *Let S be a henselian local scheme and s its closed point. The functor*

$$G \longmapsto G_s$$

induces a bijection from the set of isomorphism classes of reductive S -groups onto the set of isomorphism classes of reductive $\kappa(s)$ -groups.

⁽¹⁰⁾**N.D.E.** Cf. XXIII, Definition 5.11.

In particular, for every reductive S -group G , there exists a finite surjective étale morphism $S' \rightarrow S$ such that $G_{S'}$ is splittable.

PROOF. Using the existence of the $\acute{\text{E}}\text{p}_S(\mathcal{R})$ (Exp. XXV), one is reduced to proving that, if one puts $H = A_S(\mathcal{R})$, the canonical map

$$\text{Fib}(S, H) \rightarrow \text{Fib}(\kappa(s), H_s)$$

is bijective (and that every element of $\text{Fib}(S, H)$ has the property indicated above). Now every finite subset of H is contained in an affine open subset (this is indeed trivial for a constant group, and H is affine over a constant group); one can therefore use the result proved in the appendix (8.1). $\square$

2. Automorphisms and subgroups

Let us introduce some notation: if $H = \underline{\text{Aut}}_{S\text{-gr.}}(G)$, and if X is a subfunctor of G , one puts

$$\begin{aligned} \underline{\text{Aut}}_{S\text{-gr.}}(G, X) &= \underline{\text{Norm}}_H(X), \\ \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_X) &= \underline{\text{Centr}}_H(X). \end{aligned}$$

If Y is a second subfunctor of G , one likewise defines $\underline{\text{Aut}}_{S\text{-gr.}}(G, X, Y) = \underline{\text{Aut}}_{S\text{-gr.}}(G, X) \cap \underline{\text{Aut}}_{S\text{-gr.}}(G, Y)$, and if G' is a second S -group and X' a subfunctor of G , one denotes by $\underline{\text{Isom}}_{S\text{-gr.}}(G, X; G', X')$ the subfunctor of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$ defined by: for every $S' \rightarrow S$,

$$\begin{aligned} \underline{\text{Isom}}_{S\text{-gr.}}(G, X; G', X')(S') \\ = \{u \in \underline{\text{Isom}}_{S\text{-gr.}}(G, G')(S') \mid u(X_{S'}) = X'_{S'}\} \end{aligned}$$

and one similarly defines $\underline{\text{Isom}}_{S\text{-gr.}}(G, X, Y; G', X', Y')$, etc.⁽¹¹⁾

Proposition 2.1. *Let S be a scheme, G a reductive S -group, and T a maximal torus of G (resp. B a Borel subgroup of G , resp. $B \supset T$ a Killing pair of G). Denote by T^{ad} (resp. B^{ad}) the maximal torus (resp. Borel subgroup) of $\text{ad}(G)$ corresponding to T (resp. B):*

$$\begin{aligned} B^{\text{ad}} &\simeq B/\underline{\text{Centr}}(G) = B/\underline{\text{Centr}}(B), \\ T^{\text{ad}} &\simeq T/\underline{\text{Centr}}(G). \end{aligned}$$

Then $\underline{\text{Aut}}_{S\text{-gr.}}(G, T)$ (resp. $\underline{\text{Aut}}_{S\text{-gr.}}(G, B)$, resp. $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)$) is representable by a closed subscheme of $\underline{\text{Aut}}_{S\text{-gr.}}(G)$, smooth over S , and the exact sequence of Theorem 1.3 induces exact sequences:

$$\begin{aligned} 1 &\rightarrow \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}}) \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, T) \rightarrow \underline{\text{Autext}}(G) \rightarrow 1; \\ 1 &\rightarrow B^{\text{ad}} \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, B) \rightarrow \underline{\text{Autext}}(G) \rightarrow 1; \\ 1 &\rightarrow T^{\text{ad}} \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, B, T) \rightarrow \underline{\text{Autext}}(G) \rightarrow 1. \end{aligned}$$

PROOF. By descent of closed subschemes,⁽¹²⁾ one is immediately reduced to the case where G is pinned and $B \supset T$ is its canonical Killing pair (cf. Exp. XXII, 5.5.5 (iv)). Since the group E of 1.2 normalizes B and T , the result follows at once from the normalizer theorems in $\text{ad}(G)$ (Exp. XXII, 5.3.12 and 5.6.1). $\square$

Using now the conjugacy theorems (cf. Exp. XXIII, 5.12), and reasoning as in no. 1, one deduces:

Corollary 2.2. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point. Let $B \supset T$ (resp. $B' \supset T'$) be a Killing pair of G (resp. G').*

⁽¹¹⁾**N.D.E.** We have made the preceding definitions explicit (the original indicated “One similarly defines $\underline{\text{Aut}}_{S\text{-gr.}}(G, X, Y), \dots, \underline{\text{Isom}}_{S\text{-gr.}}(G, X; G', X'), \dots$ ”).

⁽¹²⁾**N.D.E.** Cf. SGA 1, VIII 1.9.

- (i) The S -functor $\underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')$ is representable by a smooth closed subscheme of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$ which is principal homogeneous under $\underline{\text{Aut}}_{S\text{-gr.}}(G, T)$.

Moreover, $\underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}})$ acts freely on this scheme, and one has a canonical isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T') / \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}}) \simeq \underline{\text{Isomext}}(G, G').$$

- (ii) The S -functor $\underline{\text{Isom}}_{S\text{-gr.}}(G, B; G', B')$ is representable by a smooth closed subscheme of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$ which is principal homogeneous under $\underline{\text{Aut}}_{S\text{-gr.}}(G, B)$.

Moreover, B^{ad} acts freely on this scheme and one has a canonical isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, B; G', B') / B^{\text{ad}} \simeq \underline{\text{Isomext}}(G, G').$$

- (iii) The S -functor $\underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T')$ is representable by a smooth closed subscheme of $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$, principal homogeneous under $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)$.

Moreover, T^{ad} acts freely on this scheme and one has a canonical isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T') / T^{\text{ad}} \simeq \underline{\text{Isomext}}(G, G').$$

Reasoning again as in no. 1, one deduces:

Corollary 2.3. *Let S be a scheme, G a reductive S -group, and $B \supset T$ a Killing pair of G . The functor*

$$(G', T') \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T'),$$

resp.

$$(G', B') \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, B; G', B'),$$

resp.

$$(G', B', T') \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T'),$$

is an equivalence between the category of pairs (G', T') (resp. of pairs (G', B') , resp. of triples (G', B', T')), where G' is a form of G and T' a maximal torus of G' (resp. B' a Borel subgroup of G' , resp. $B' \supset T'$ a Killing pair of G'), and the category of principal homogeneous bundles under the S -group H , where $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, T)$ (resp. $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, B)$, resp. $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)$).

Moreover, every principal homogeneous sheaf under H is representable and quasi-isotrivial, so that one has

$$H^1(S, H) = H^1_{\text{ét}}(S, H) = \text{Fib}(S, H).$$

Remark 2.4. Under the conditions of 2.2, the morphism denoted set-theoretically by $u \longmapsto u(T)$ (resp. $u \longmapsto u(B)$, resp. $u \longmapsto (u(B), u(T))$) induces an isomorphism

$$\underline{\text{Isom}}_{S\text{-gr.}}(G, G') / \underline{\text{Aut}}_{S\text{-gr.}}(G, T) \simeq \underline{\text{Tor}}(G'),$$

$$\text{resp. } \underline{\text{Isom}}_{S\text{-gr.}}(G, G') / \underline{\text{Aut}}_{S\text{-gr.}}(G, B) \simeq \underline{\text{Bor}}(G'),$$

$$\text{resp. } \underline{\text{Isom}}_{S\text{-gr.}}(G, G') / \underline{\text{Aut}}_{S\text{-gr.}}(G, B, T) \simeq \underline{\text{Kil}}(G').$$

The proof is immediate: it suffices to give it locally for (fpqc), so one may suppose $G \simeq G'$, and one is reduced to Exp. XXII, 5.8.3 (iii).

Remark 2.5. The preceding results can immediately be interpreted in terms of restriction of the structure group: if G' is a form of G , corresponding to the principal bundle $\underline{\text{Isom}}_{S\text{-gr.}}(G, G')$, giving a restriction of the structure group of this bundle to $\underline{\text{Aut}}_{S\text{-gr.}}(G, T)$ amounts to giving a maximal torus T' of G' , the bijections suggested above being that of 2.4 on the one hand, and the map $T' \longmapsto \underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')$ on the other hand. Likewise for Borel subgroups and Killing pairs.

Proposition 2.6. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point, and T (resp. T') a maximal torus of G (resp. G'). Then T^{ad} acts freely on*

$\underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')$, and the quotient

$$P = \underline{\text{Isom}}_{S\text{-gr.}}(G, T; G', T')/T^{\text{ad}}$$

is representable; it is a principal homogeneous bundle under

$$A = \underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}},$$

where A is representable by a twisted constant S -scheme, an extension of $\underline{\text{Autext}}(G)$ by $\text{W}_{\text{ad}(G)}(T^{\text{ad}}) = \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}})/T^{\text{ad}}$. Moreover, if one lets A act on T in the evident way, the bundle associated to P is nothing other than T' .⁽¹³⁾

PROOF. The first part of the proposition follows at once from the preceding results. To prove the second, one observes that there is an evident morphism $P \times_S T \rightarrow T'$ (defined by $(u, t) \mapsto u(t)$); to prove that after passage to the quotient by A it induces an isomorphism, one may once more suppose $(G, T) \simeq (G', T')$, in which case it is immediate. $\square$

In an entirely similar way, one has:

Proposition 2.7. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point, and $B \supset T$ (resp. $B' \supset T'$) a Killing pair of G (resp. G'). If one lets $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)/T^{\text{ad}} \simeq \underline{\text{Autext}}(G)$ act on T in the evident way, the bundle associated to $\underline{\text{Isomext}}(G, G')$ is nothing other than T' .*

Corollary 2.8. *Let G and G' be two reductive S -groups which are inner twisted forms of one another; let $B \supset T$ (resp. $B' \supset T'$) be a Killing pair of G (resp. G'). Then T and T' are isomorphic.⁽¹⁴⁾*

Remark 2.9. It is not true in general that B and B' are isomorphic; they are, however, inner twisted forms of one another (cf. no. 5).

One can develop “ $\underline{\text{Isomint}}$ ”⁽¹⁵⁾ variants of the preceding results. Let us mention one:

Proposition 2.10. *Let S be a scheme and G, G' two reductive S -groups of the same type at every point, and $B \supset T$ (resp. $B' \supset T'$) a Killing pair of G (resp. G'). Let $u \in \underline{\text{Isomext}}(G, G')(S)$; consider*

$$\begin{aligned} & \underline{\text{Isomint}}_u(G, B, T; G', B', T') \\ &= \underline{\text{Isom}}_{S\text{-gr.}}(G, B, T; G', B', T') \cap \underline{\text{Isomint}}_u(G, G'). \end{aligned}$$

It is a smooth affine S -scheme which is a principal homogeneous bundle under T^{ad} . In particular, $\underline{\text{Isomint}}_u(G, B, T; G', B', T')(S) \neq \emptyset$ if and only if the corresponding element of $H^1(S, T^{\text{ad}})$ is zero.

To finish this section, let us prove two results which will be useful to us in what follows:

Proposition 2.11. *Let S be a scheme, G a reductive S -group, and T a maximal torus of G . The evident morphism*

$$T^{\text{ad}} \xrightarrow{\sim} \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_T)$$

is an isomorphism.

This follows from the preceding statements; moreover, a direct proof has been given in the course of the proof of 1.5.2.

⁽¹³⁾**N.D.E.** Thus, if P has a section, then T and T' are isomorphic; in particular T'_P and T_P are isomorphic. See the addition 4.5.1 where this observation is developed in the case where $G' = \acute{\text{E}}\text{p}_S(\mathcal{A})$.

⁽¹⁴⁾**N.D.E.** By 2.7, since $\underline{\text{Isomext}}(G, G')$ has a section (cf. the preceding N.D.E.).

⁽¹⁵⁾**N.D.E.** Recall that $\underline{\text{Isomint}}_u(G, G')$ is defined in 1.11.

Corollary 2.12. *Under the preceding conditions, there exists an equivalence between the category of pairs (G', f) , where G' is a form of G and f an isomorphism of T onto a maximal torus of G' , and the category of principal homogeneous bundles under T^{ad} .*

Corollary 2.13. *If $H^1(S, T^{\text{ad}}) = 0$, and if G' is a form of G possessing a maximal torus isomorphic to T , then G' is isomorphic to G .*

Corollary 2.14. *Let S be a scheme such that $\text{Pic}(S) = 0$, and G a reductive S -group of constant type. For G to be splittable, it is necessary and sufficient that G possess a split maximal torus.*

Let G be a reductive group and $\text{rad}(G)$ its radical (Exp. XXII, 4.3.9); since $\text{rad}(G)$ is central and characteristic in G , one has a canonical morphism

$$q : \underline{\text{Autext}}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(\text{rad}(G)).$$

Proposition 2.15. *Let G be a reductive S -group. The following sequence is exact:*

$$1 \longrightarrow \text{ad}(G) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_{\text{rad}(G)})^p \longrightarrow \underline{\text{Autext}}(G) \xrightarrow{q} \underline{\text{Aut}}_{S\text{-gr.}}(\text{rad}(G))$$

and $\text{Ker}(q) = \text{Im}(p)$ is an open and closed subscheme of $\underline{\text{Autext}}(G)$, finite over S .

PROOF. One may suppose G split. The first assertion is immediate; the second follows from Exp. XXI, 6.7.5 and 6.7.7. $\square$

Putting $H = \underline{\text{Aut}}_{S\text{-gr.}}(G, \text{id}_{\text{rad}(G)})$, one deduces:

Corollary 2.16. *There exists an equivalence between the category of pairs (G', f) , where G' is a form of G and f an isomorphism of $\text{rad}(G)$ onto the radical of G' , and the category of principal bundles under a certain S -group scheme H , where H is such that there exists an exact sequence*

$$1 \longrightarrow \text{ad}(G) \longrightarrow H \longrightarrow F \longrightarrow 1,$$

where the S -group F is étale and finite over S .

3. Dynkin scheme of a reductive group. Quasi-split groups

3.1. — Recall (Exp. XXI, 7.4.1) that a *Dynkin diagram* is a finite set equipped with the structure defined by a set of pairs of distinct elements (bonds) and a map into $\{1, 2, 3\}$ (lengths). To each reduced pinned root datum $\mathcal{R}$ is associated a Dynkin diagram $\Delta(\mathcal{R})$, whose underlying set is the set of simple roots.

3.2. — Let S be a scheme. A *Dynkin S -scheme* is a finite twisted constant S -scheme Δ , equipped with the structure defined by a subscheme L of $\Delta \times_S \Delta$ having empty intersection with the diagonal, and a morphism $\Delta \longrightarrow \{1, 2, 3\}_S$. If S' is a connected⁽¹⁶⁾ S -scheme, $\Delta(S')$ is naturally equipped with the structure of a Dynkin diagram.

One immediately defines the following notions: isomorphism of two Dynkin schemes, extension of the base of a Dynkin scheme, constant Dynkin scheme associated to a Dynkin diagram.

Every descent datum on a Dynkin scheme for the étale topology is effective.

3.3. — One proposes to associate to every reductive S -group G a Dynkin S -scheme. Suppose first that G is *splittable* over S ; for every pinning $\mathcal{E}$ of G , denote by $\Delta(\mathcal{E})$ the constant Dynkin scheme associated to the pinned root datum defined by $\mathcal{E}$; if $\mathcal{E}$ and $\mathcal{E}'$ are two pinnings of G , there exists by 1.5 a *unique inner automorphism* of G over S taking $\mathcal{E}$ to $\mathcal{E}'$; this automorphism of G defines an isomorphism $a_{\mathcal{E}\mathcal{E}'} : \Delta(\mathcal{E}) \xrightarrow{\sim} \Delta(\mathcal{E}')$; the $a_{\mathcal{E}\mathcal{E}'}$ clearly form a transitive system, so that one can identify the $\Delta(\mathcal{E})$ (i.e. take the inductive limit); the result

⁽¹⁶⁾**N.D.E.** We have added the hypothesis that S' be connected.

is a constant Dynkin scheme denoted by $\underline{\text{Dyn}}(G)$. If now G is an *arbitrary* reductive S -group, there exists a covering family for the étale topology $\{S_i \rightarrow S\}$ such that G_{S_i} is splittable. Reasoning as before, one therefore has a canonical descent datum on the $\underline{\text{Dyn}}(G_{S_i})$, allowing one to construct by descent a Dynkin S -scheme $\underline{\text{Dyn}}(G)$.

3.4. — This construction satisfies the following properties (which, moreover, essentially characterize it):

- (i) To every reductive S -group is associated a Dynkin scheme $\underline{\text{Dyn}}(G)$; to every isomorphism $u : G \xrightarrow{\sim} G'$ is functorially associated an isomorphism $\underline{\text{Dyn}}(u) : \underline{\text{Dyn}}(G) \xrightarrow{\sim} \underline{\text{Dyn}}(G')$.
- (ii) If S' is an S -scheme and G a reductive S -group, one has

$$\underline{\text{Dyn}}(G \times_S S') \simeq \underline{\text{Dyn}}(G) \times_S S'.$$

- (iii) If $\mathcal{E}$ is a pinning of G , defining the pinned root datum with Dynkin diagram Δ , one has

$$\underline{\text{Dyn}}(G) \simeq \Delta_S.$$

- (iv) If u is an inner automorphism of G , $\underline{\text{Dyn}}(u)$ is the identity automorphism of $\underline{\text{Dyn}}(G)$.

3.5. — Let S be a scheme and G a reductive S -group. It is clear that the functor $\underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G))$ of automorphisms of $\underline{\text{Dyn}}(G)$ for the Dynkin scheme structure is representable by a finite twisted constant S -scheme. By 3.4 (i) and (ii), one has a canonical morphism

$$\underline{\text{Aut}}_{S\text{-gr.}}(G) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G)),$$

which, by (iv), factors through a morphism

$$\underline{\text{Autext}}(G) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G)).$$

More generally, if G and G' are two reductive S -groups, one has a canonical morphism

$$\underline{\text{Isomext}}(G, G') \rightarrow \underline{\text{Isom}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G), \underline{\text{Dyn}}(G'));$$

in particular, if G' is an inner twisted form of G (1.11), the Dynkin schemes $\underline{\text{Dyn}}(G)$ and $\underline{\text{Dyn}}(G')$ are isomorphic.

3.6. — If G is semisimple (resp. adjoint or simply connected), the morphism

$$\underline{\text{Autext}}(G) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\underline{\text{Dyn}}(G))$$

is a monomorphism (resp. an isomorphism). Indeed, one may suppose G pinned and one is reduced to the corresponding result for reduced pinned root data (cf. Exp. XXI, 7.4.5).

One has an analogous result for the $\underline{\text{Isom}}$; it follows in particular that two adjoint (resp. simply connected) semisimple S -groups are inner twisted forms of one another if and only if their Dynkin schemes are isomorphic.

3.7. — One can give a different construction of the Dynkin scheme associated to a reductive group. Let $\mathcal{R}$ be a reduced pinned root datum and G a reductive S -group of type $\mathcal{R}$; denote by $\Delta(\mathcal{R})$ the Dynkin diagram defined by the root datum $\mathcal{R}$. One has (3.5) a canonical morphism

$$A_S(\mathcal{R}) = \underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{E}}p_S(\mathcal{R})) \rightarrow \underline{\text{Aut}}_{\underline{\text{Dyn}}}(\Delta(\mathcal{R})_S).$$

The reductive S -group G corresponds (1.17) to a bundle $\underline{\text{Isom}}_{S\text{-gr.}}(\dot{\text{E}}p_S(\mathcal{R}), G)$, principal homogeneous under $A_S(\mathcal{R})$. The associated bundle under $\underline{\text{Aut}}_{\underline{\text{Dyn}}}(\Delta(\mathcal{R})_S)$ corresponds to a form over S of $\Delta(\mathcal{R})_S$: it is $\underline{\text{Dyn}}(G)$; in other words, this associated bundle is nothing other than $\underline{\text{Isom}}_{\underline{\text{Dyn}}}(\Delta(\mathcal{R})_S, \underline{\text{Dyn}}(G))$. In this latter form the proof is immediate.

3.8. — **Dynkin scheme and Killing pairs.** Let S be a scheme, G a reductive S -group, and $B \supset T$ a Killing pair of G . There exists a canonical morphism

$$i : \underline{\text{Dyn}}(G) \rightarrow \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$$

which identifies $\underline{\text{Dyn}}(G)$ with the “scheme of simple roots of B relative to T ”; this morphism is immediately defined by descent from the pinned case. Observe moreover that the datum of T and i allows one to reconstruct B (“one-to-one correspondence between systems of simple roots and systems of positive roots”).

It follows from the preceding description of $\mathcal{D} = \underline{\text{Dyn}}(G)$ that there exists a *canonical* root of $B_{\mathcal{D}}$ relative to $T_{\mathcal{D}}$: this root $\alpha_{\mathcal{D}}$ is the image under $i(\mathcal{D})$ of the identity morphism of $\mathcal{D}$. One deduces a *canonical invertible $\mathcal{O}_{\mathcal{D}}$ -module $\mathfrak{g}^{\mathcal{D}}$* :

$$\mathfrak{g}^{\mathcal{D}} = (\mathfrak{g} \otimes_S \mathcal{O}_{\mathcal{D}})^{\alpha_{\mathcal{D}}}.$$

In the pinned case one has

$$\mathcal{D} = \coprod_{\alpha \in \Delta} S_{\alpha},$$

where each S_{α} is a copy of S , and $\mathfrak{g}^{\mathcal{D}}$ is then the $\mathcal{O}_{\mathcal{D}}$ -module which induces $\mathfrak{g}^{\alpha}$ on S_{α} , for every $\alpha \in \Delta$.

3.9. — Quasi-pinnings. Quasi-pinned groups. If G is a reductive S -group, one calls a *quasi-pinning* of G the datum:

- (i) of a Killing pair $B \supset T$ of G ,
- (ii) of a section $X \in \Gamma(\underline{\text{Dyn}}(G), \mathfrak{g}^{\mathcal{D}})^{\times}$.

One says that a reductive S -group is *quasi-splittable* if it possesses a quasi-pinning. One calls a *quasi-pinned group* a reductive group equipped with a quasi-pinning.

Let $B \supset T$ be a Killing pair of the reductive S -group G ; then G admits a quasi-pinning relative to this Killing pair if and only if $\mathfrak{g}^{\mathcal{D}}$ has a section nonzero at every point, i.e. if the element of $\text{Pic}(\underline{\text{Dyn}}(G))$ defined by $\mathfrak{g}^{\mathcal{D}}$ is zero. Suppose in particular that S is semi-local; then $\underline{\text{Dyn}}(G)$ is also semi-local, hence $\text{Pic}(\underline{\text{Dyn}}(G)) = 0$. One deduces:

Proposition 3.9.1. *Let S be a semi-local scheme and G a reductive S -group. For G to be quasi-splittable, it is necessary and sufficient that it possess a Borel subgroup.*

PROOF. ⁽¹⁷⁾Indeed, S is affine, hence, by the first assertion of Exp. XXII, 5.9.7, if G possesses a Borel subgroup B , it also possesses a Killing pair $B \supset T$. Then, since $\text{Pic}(\mathcal{D}) = 0$, $\mathfrak{g}^{\mathcal{D}}$ has an everywhere nonzero section X . $\square$

Let again A be a semi-local ring and $S = \text{Spec}(A)$; observe now that for every reductive S -group G the morphism $\underline{\text{Bor}}(G) \rightarrow S$ is surjective (since $G_{\bar{s}}$ possesses Borel subgroups, for every $s \in S$) and smooth and projective (Exp. XXII, 5.8.3), hence has sections after a finite surjective étale extension of the base.⁽¹⁸⁾ One deduces the

⁽¹⁷⁾N.D.E. We have given the reference to Exp. XXII, 5.9.7 in detail.

⁽¹⁸⁾N.D.E. The original referred to EGA IV, § 24, which did not appear. One must use “Bertini’s theorem”. Let us give the argument in detail, as indicated to us by O. Gabber. Let $X \rightarrow S$ be a surjective, smooth and projective morphism; replacing S by a connected component, one may suppose that X/S is everywhere of relative dimension d (cf. EGA IV₄, 17.10.2). One may also suppose that X is a closed subscheme of a projective space $\mathbf{P}_S^n = \mathbf{P}(E)$, where E is a free A -module of rank $n+1$ (cf. EGA II, 5.3.3). Let $s_1, \dots, s_r$ be the closed points of S ; by Bertini’s theorem (see for example [Jou83], I 6.10), there exists an open subset U of the product $P = \mathbf{P}(E^*)^d$, with nonempty fibers, such that for every point $u = (f_1, \dots, f_d)$ of U_{s_i} , the intersection of $X_{\kappa(u)}$ with the d hyperplanes of $\mathbf{P}_{\kappa(u)}^n$ defined by u is étale over $\kappa(u)$. After shrinking U , one may suppose that U is the complement in affine space $\mathbf{A}_S^{nd}$ of the zero locus $\mathcal{V}(Q)$ of a certain polynomial Q of degree $m > 0$. One then sees easily (by induction on the number of variables) that $\mathcal{V}(Q)$ cannot contain all the rational points of $\mathbf{A}_{s_i}^{nd}$ if $|\kappa(s_i)| > m$. Since the morphism $A \rightarrow \prod_i \kappa(s_i)$ is surjective, one can find a monic polynomial $R \in A[X]$ of degree m whose image in $\kappa(s_i)[X]$ is irreducible if $\kappa(s_i)$ is finite, and has m distinct roots if $\kappa(s_i)$ is infinite. Put $A' = A[X]/(R)$ and $S' = \text{Spec}(A')$; then $S' \rightarrow S$ is étale, finite and surjective, and the residue field at each of its closed points $s'_1, \dots, s'_t$ has cardinality $\geq 2^m$. Then U has a rational point u_i over each closed point of S' , and since $A' \rightarrow \prod_i \kappa(s_i)$ is surjective, these lift to a section u of $P_{S'}$. Denote by Z the intersection of $X_{S'}$ with the d hyperplanes of $\mathbf{P}_{S'}^n$ defined by u , and by V the open subset of Z consisting of points where Z is étale over S' . By EGA IV₃, 11.3.8, V contains the fibers $Z_{s'_i}$ for every i ; since $\pi : Z \rightarrow S'$ is proper, it follows that the closed subset $\pi(Z - V)$ is empty, whence $V = Z$.

Corollary 3.9.2. *Let S be a semi-local scheme and G a reductive S -group. There exists an étale, finite and surjective morphism $S' \rightarrow S$ such that $G_{S'}$ is quasi-splittable.*

Remark 3.9.3. Under the preceding conditions, let T be a maximal torus of G (cf. Exp. XIV, 3.20); then one may suppose in addition that $G_{S'}$ is quasi-splittable relative to $T_{S'}$: it suffices to apply the preceding reasoning to the “scheme of Borel subgroups containing T ”, which is finite and étale over S (Exp. XXII, 5.5.5 (ii)).

3.10. — Let $\mathcal{E}$ and $\mathcal{E}'$ be two quasi-pinnings of the reductive S -group G . There exists a *unique* inner automorphism of G taking $\mathcal{E}$ to $\mathcal{E}'$. Indeed, one is immediately reduced to the split case, where the assertion has already been proved (1.5; indeed, it suffices to observe that for an inner automorphism of G it amounts to the same thing to respect a pinning or the underlying quasi-pinning). One concludes, as in no. 1, that a quasi-pinning of the reductive S -group G defines a splitting h of the exact sequence:

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \xleftarrow[\substack{h \\ p}]{\substack{h \\ p}} \underline{\mathrm{Autext}}(G) \longrightarrow 1,$$

the image of h being the subgroup of $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ which leaves the quasi-pinning invariant.

Likewise, if G and G' are two quasi-pinned S -groups, one naturally defines the subfunctor

$$\underline{\mathrm{Isom}}_{S\text{-gr. q-pin.}}(G, G')$$

of $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$; the projection from $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ onto $\underline{\mathrm{Isomext}}(G, G')$ induces an isomorphism

$$(*) \quad \underline{\mathrm{Isom}}_{S\text{-gr. q-pin.}}(G, G') \xrightarrow{\sim} \underline{\mathrm{Isomext}}(G, G').$$

Theorem 3.11. *Let S be a scheme, $\mathcal{R}$ a reduced pinned root datum such that $\acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})$ exists (cf. Exp. XXV), and E the group of its automorphisms. Consider the following three categories:*

- (i) *The category $\underline{\mathrm{Rev}}$ of principal Galois coverings of S with group E (the morphisms are the isomorphisms of E -bundles).*
- (ii) *The category $\underline{\mathrm{Redext}}$ whose objects are the reductive S -groups of type $\mathcal{R}$ (Exp. XXII, 2.7), the morphisms from G to G' being the elements of $\underline{\mathrm{Isomext}}(G, G')(S)$.*
- (iii) *The category $\underline{\mathrm{Qép}}$ of quasi-pinned reductive S -groups of type $\mathcal{R}$ (the morphisms are the isomorphisms respecting the quasi-pinnings).*

These three categories are equivalent: more precisely, one has a diagram of functors, commutative up to functorial isomorphisms

$$\begin{array}{ccc} \underline{\mathrm{Rev}} & \xrightarrow{\quad \mathrm{qép} \quad} & \underline{\mathrm{Qép}} \\ & \swarrow \mathrm{rev} \quad \searrow i & \\ & \underline{\mathrm{Redext}} & \end{array} .$$

We shall describe these three functors below, leaving to the reader the task of verifying the commutativity of the diagram.⁽¹⁹⁾

3.11.1. — The functor i . It is the evident functor: $i(G) = G$, $i(f) = \text{image of } f \text{ under the isomorphism of 3.10 } (*)$.

3.11.2. — The functor $\mathrm{qép}$. Let S' be a principal Galois covering of S with group E . Let $(G, T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ be a pinning of $G = \acute{\mathrm{E}}\mathrm{p}_S(\mathcal{R})$ and let $(X'_\alpha)_{\alpha \in \Delta}$ be the corresponding

Then $Z \rightarrow S'$ is surjective, étale and proper, hence finite, as is the composite $Z \rightarrow S$, and this supplies the desired section of $X \rightarrow S$.

⁽¹⁹⁾**N.D.E.** In 3.11.4 we have sketched the verification that the functors i , rev and $\mathrm{qép}$ are fully faithful, and that the composite $\mathrm{rev} \circ i \circ \mathrm{qép}$ is isomorphic to the identity functor of $\underline{\mathrm{Rev}}$.

pinning of $G' = G_{S'} = \dot{\text{Ép}}_{S'}(\mathcal{R})$. By Exp. XXIII 5.5 bis, there exists a unique group morphism

$$\theta : E \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{Ép}}_S(\mathcal{R})) = A(\mathcal{R})(S)$$

such that, for every $h \in E$, $\theta(h)$ induces $D_S(h)$ on T and $\text{Lie}(\theta(h))(X_\alpha) = X_{h(\alpha)}$ for every $\alpha \in \Delta$. Taking account of the action of E on S' , one thus obtains an action of E on G' , compatible with the action of E on S' and respecting the canonical quasi-pinning of G' .⁽²⁰⁾ Since $S' \longrightarrow S$ is covering for the (fpqc) topology, it is a morphism of effective descent for the fibered category of affine morphisms, and one denotes by

$$\text{qép}(S') = \text{Q-Ép}_{S'/S}(\mathcal{R})$$

the *quasi-pinned* S -group obtained by Galois descent.⁽²¹⁾

3.11.3. — The functor rev. Let G be a reductive S -group of type $\mathcal{R}$. One puts

$$\text{rev}(G) = \underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), G),$$

it is a principal homogeneous bundle under E_S (cf. 1.10 and 1.19), i.e. an object of Rev.

3.11.4. — ⁽²²⁾It is clear, by the isomorphism 3.10 (*), that the functor i is fully faithful. Moreover, if G_0, G, G' are reductive S -groups, the natural morphism

$$\underline{\text{Isom}}(G_0, G) \times_S \underline{\text{Isom}}(G, G') \longrightarrow \underline{\text{Isom}}(G_0, G')$$

induces a morphism

$$(*) \quad \underline{\text{Isomext}}(G_0, G) \times_S \underline{\text{Isomext}}(G, G') \longrightarrow \underline{\text{Isomext}}(G_0, G')$$

since, for every $S' \longrightarrow S$, if $g_0 \in G_0(S')$, $g \in G(S')$, $u \in \underline{\text{Isom}}(G_0, G)(S')$, and $v \in \underline{\text{Isom}}(G, G')(S')$, one has the equality $v \circ \text{int}(g) \circ u \circ \text{int}(g_0) = v \circ u \circ \text{int}(u^{-1}(g)g_0)$. Taking $G_0 = \dot{\text{Ép}}_S(\mathcal{R})$, one obtains a map

$$\begin{aligned} \underline{\text{Isomext}}(G, G')(S) &\longrightarrow \underline{\text{Hom}}_{\text{Rev}}(\underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), G), \\ &\quad \underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), G')) \end{aligned}$$

and to show that it is a bijection, one may suppose, by (fpqc) descent, that G and G' are pinned, in which case it is evident. Likewise, if S_1, S_2 are two objects of Rev and if one denotes by $S_i \times^E \dot{\text{Ép}}_S(\mathcal{R})$ the S -group $\text{qép}(S_i)$ obtained by twisting $\dot{\text{Ép}}_S(\mathcal{R})$ by the E -torsor S_i (cf. N.D.E. (21)), one has a natural map

$$\begin{aligned} \underline{\text{Hom}}_{\text{Rev}}(S_1, S_2) &\longrightarrow \underline{\text{Hom}}_{\text{Qép}}(S_1 \times^E \dot{\text{Ép}}_S(\mathcal{R}), \\ &\quad S_2 \times^E \dot{\text{Ép}}_S(\mathcal{R})), \end{aligned}$$

and to show that it is a bijection, one may suppose, by (fpqc) descent, that S_1 and S_2 are trivial E -bundles, in which case it is evident.

⁽²⁰⁾**N.D.E.** We have expanded the original in the above, to show that the action of E on G' is obtained by combining the natural action of E on $\dot{\text{Ép}}_{S'}(\mathcal{R})$ and the given action on S' . The canonical Killing pair of G' is preserved by this action, as is the quasi-pinning given by the element $\tilde{X}'$ of $\Gamma(\Delta_{S'}, \text{Lie}(G'/S')^{\Delta_{S'}})$, equal to X'_α on the copy of S' indexed by α (indeed, since E acts on $\Delta_{S'}$ by permuting the copies of S' , one has $h(\tilde{X}') = \tilde{X}'$ for every $h \in E$).

⁽²¹⁾**N.D.E.** See, for example, TDTE I, p. 22, Example 1. One can also describe $\text{qép}(S')$ as the twist of $G = \dot{\text{Ép}}_S(\mathcal{R})$ by the E -torsor S'/S , i.e. the (fpqc) quotient sheaf of $S' \times_S G$ by the right action of E defined by $(s', g) \cdot h = (s'h, h^{-1}(g))$. Since G is affine over S and E acts on G by group automorphisms, this sheaf is representable by an S -group $G^\sharp$, which is a “twisted” form of G , and $\mathcal{D}^\sharp = \underline{\text{Dyn}}(G^\sharp)$ is the twist of the constant Dynkin scheme Δ_S by the torsor S'/S . Since E normalizes B and T , one similarly obtains a pair $(B^\sharp, T^\sharp)$ which is a Killing pair of $G^\sharp$. Moreover, let $\mathfrak{g} = \text{Lie}(G/S)$ and $\mathfrak{g}^\sharp = \text{Lie}(G^\sharp/S)$; for every $U \longrightarrow S$, the sections of $\mathfrak{g}^\sharp$ over U are the E -equivariant S -morphisms $U \times_S S' \longrightarrow \mathbf{W}(\mathfrak{g})$. Since $\mathcal{D}^\sharp \times_S S'$ is E -isomorphic to $\Delta \times S'$, equipped with the action $(\alpha, s') \cdot h = (h^{-1}(\alpha), s'h)$, one obtains that the morphism given by $(\alpha, s') \longmapsto X_\alpha$ is a section of $\mathfrak{g}^\sharp$ over $\mathcal{D}^\sharp$ which is a quasi-pinning, i.e. an everywhere nonzero section of $(\mathfrak{g}^\sharp \otimes_{\mathcal{O}_S} \mathcal{O}_{\mathcal{D}^\sharp})^{\mathcal{D}^\sharp}$ (cf. 3.8).

⁽²²⁾**N.D.E.** We have added this paragraph.

Thus the three functors of the diagram are fully faithful. Moreover, for every object S' of $\underline{\text{Rev}}$ one has a natural morphism

$$S' \longrightarrow \underline{\text{Isomext}}(\dot{\text{Ép}}_S(\mathcal{R}), S' \times^E \dot{\text{Ép}}_S(\mathcal{R}))$$

and to see that it is an isomorphism, one may suppose, by (fpqc) descent, that S' is a trivial E -bundle, in which case it is evident.

One therefore has an isomorphism of functors $\text{Id}_{\underline{\text{Rev}}} \longrightarrow \text{rev} \circ i \circ \text{qép}$. Since the functor rev is fully faithful one therefore obtains, for every reductive S -group H (not necessarily quasi-split), a bijection

$$\underline{\text{Isomext}}(\text{Q-}\dot{\text{Ép}}_{\text{rev}(H)/S}(\mathcal{R}), H)(S) \simeq \text{Aut}_{\underline{\text{Rev}}}(\text{rev}(H))$$

functorial in H . In particular, the identity morphism of $\text{rev}(H)$ corresponds to a “canonical” element u_0 of

$$\underline{\text{Isomext}}(\text{Q-}\dot{\text{Ép}}_{\text{rev}(H)/S}(\mathcal{R}), H)(S);$$

moreover, every

$$u \in \underline{\text{Isomext}}(\text{Q-}\dot{\text{Ép}}_{\text{rev}(H)/S}(\mathcal{R}), H)(S)$$

corresponds to an automorphism ϕ_u of $\text{rev}(H)$, and $\text{qép}(\phi_u)$ is the unique automorphism of the pinned S -group $\text{Q-}\dot{\text{Ép}}_{\text{rev}(H)/S}(\mathcal{R})$ such that $u_0 \circ \text{qép}(\phi_u) = u$. We have thus proved Theorem 3.11. Let us develop one of the corollaries of 3.11:

Corollary 3.12. *For every reductive S -group G , there exists a quasi-pinned S -group $G_{q\text{-pin}}$, and an “outer isomorphism” $u \in \underline{\text{Isomext}}(G_{q\text{-pin}}, G)(S)$. The pair $(G_{q\text{-pin}}, u)$ is unique up to a unique isomorphism.*

Indeed, one may suppose G of constant type $\mathcal{R}$, and one takes $G_{q\text{-pin}} = \text{Q-}\dot{\text{Ép}}_{\text{rev}(G)/S}(\mathcal{R})$.

3.12.1. — Observe that the datum of the pair $(G_{q\text{-pin}}, u)$ allows one to define canonically the S -scheme (cf. 1.11)

$$Q = \underline{\text{Isomint}}(G_{q\text{-pin}}, G),$$

⁽²³⁾ which is a principal homogeneous bundle under $\text{ad}(G_{q\text{-pin}})$, and whose datum “is equivalent” to that of the inner twisted form G of $G_{q\text{-pin}}$. Moreover Q is nothing other than the “scheme of quasi-pinnings of G ”, a definition which accounts for its structure of principal homogeneous bundle (on the left) under $\text{ad}(G)$ (3.10)—compare with 1.20.

Proposition 3.13. *Let S be a scheme, G an adjoint (resp. simply connected) semisimple S -group, $B \supset T$ a Killing pair of G , and $\underline{\text{Dyn}}(G)$ the Dynkin S -scheme of G . There exists a canonical isomorphism of S -group schemes*

$$T \xrightarrow{\sim} \prod_{\underline{\text{Dyn}}(G)/S} \mathbf{G}_{\text{m}, \underline{\text{Dyn}}(G)}.$$

(Recall (cf. Exp. II, § 1) that the right-hand side is by definition the S -functor which associates to $S' \longrightarrow S$ the group $\mathbf{G}_{\text{m}}(\underline{\text{Dyn}}(G) \times_S S')$, or, equivalently, $\mathbf{G}_{\text{m}}(\underline{\text{Dyn}}(G_{S'}))$.)

FIRST PROOF. For simplicity let us give it in the adjoint case. Consider the composite morphism

$$T \longrightarrow \prod_{\mathcal{D}/S} T_{\mathcal{D}} \longrightarrow \prod_{\mathcal{D}/S} \mathbf{G}_{\text{m}, \mathcal{D}},$$

where the first morphism is the canonical morphism and the second is $\prod_{\mathcal{D}/S} \alpha_{\mathcal{D}}$ (we have put $\mathcal{D} = \underline{\text{Dyn}}(G)$). To verify that this morphism is an isomorphism, one may suppose G split; now, in this case, it is nothing other than the morphism $T \longrightarrow (\mathbf{G}_{\text{m}, S})^{\Delta}$ with components α , for $\alpha \in \Delta$, and the latter is an isomorphism (Exp. XXII, 4.3.8). $\square$

⁽²³⁾**N.D.E.** We write $\underline{\text{Isomint}}(G_{q\text{-pin}}, G)$ instead of $\underline{\text{Isomint}}_u(G_{q\text{-pin}}, G)$ (cf. 1.11), since the pair $(G_{q\text{-pin}}, u)$ is unique up to a unique isomorphism.

SECOND PROOF. By 2.8, 3.5 and 3.11, one may suppose that $G = \mathbf{Q}\text{-}\dot{\text{E}}\text{p}_{S'/S}(\mathcal{R})$, T being the canonical maximal torus. Over S' , one has by Exp. XXII 4.3.8 an isomorphism $T_{S'} \xrightarrow{\sim} (\mathbf{G}_{m,S'})^\Delta$, defined by the simple roots (resp. the simple coroots). The group E acts on the right-hand side by permuting Δ . Now the bundle associated to S'/S by $E \rightarrow \text{Aut}(\Delta)$ is $\underline{\text{Dyn}}(G)$ (3.7), and one concludes at once. $\square$

Using Prop. 8.4 of the appendix, one derives:

Corollary 3.14. *Under the preceding conditions, one has*

$$H^1(S, T) \xrightarrow{\sim} H^1(\underline{\text{Dyn}}(G), \mathbf{G}_m) \xrightarrow{\sim} \text{Pic}(\underline{\text{Dyn}}(G)).$$

In particular, $H^1(S, T) = 0$ when S is semi-local.

Remark 3.15. Let S be a scheme, G (resp. G') a reductive S -group, $B \supset T$ (resp. $B' \supset T'$) a Killing pair of G (resp. G'), and $u \in \underline{\text{Isomext}}(G, G')(S)$. Put (cf. 2.10)

$$P = \underline{\text{Isomint}}_u(G, B, T; G', B', T');$$

it is a principal homogeneous bundle under T^{ad} (by $(f, t) \mapsto f \circ \text{int}(t)$).

Let, on the other hand, $\mathcal{D} = \underline{\text{Dyn}}(G) = \underline{\text{Dyn}}(G')$ (identified by means of u (3.5)), and let $L = \underline{\text{Isom}}_{\mathcal{D}}(\mathfrak{g}^{\mathcal{D}}, \mathfrak{g}'^{\mathcal{D}})$ be the principal homogeneous bundle under $\mathbf{G}_{m,\mathcal{D}}$ defined by the invertible $\mathcal{O}_{\mathcal{D}}$ -module

$$\underline{\text{Hom}}_{\mathcal{O}_{\mathcal{D}}}(\mathfrak{g}^{\mathcal{D}}, \mathfrak{g}'^{\mathcal{D}}) = \mathfrak{g}'^{\mathcal{D}} \otimes (\mathfrak{g}^{\mathcal{D}})^{\vee}.$$

Each $f \in P(S')$ defines an isomorphism of $\mathfrak{g}^{\mathcal{D}} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ onto $\mathfrak{g}'^{\mathcal{D}} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, whence a canonical morphism

$$P \longrightarrow \prod_{\mathcal{D}/S} L.$$

This morphism is an *isomorphism*, compatible with the isomorphism of acting groups

$$T^{\text{ad}} \xrightarrow{\sim} \prod_{\mathcal{D}/S} \mathbf{G}_{m,\mathcal{D}}$$

defined above. Indeed, it suffices to verify it in the case where the two groups are pinned, where it is easy.

It follows in particular that in the isomorphism

$$H^1(S, T^{\text{ad}}) \xrightarrow{\sim} \text{Pic}(\underline{\text{Dyn}}(G))$$

of 3.14, the class of the bundle P is taken to $\text{cl}(\mathfrak{g}'^{\mathcal{D}}) - \text{cl}(\mathfrak{g}^{\mathcal{D}})$. The bundle P is therefore trivial if and only if $\mathfrak{g}'^{\mathcal{D}}$ and $\mathfrak{g}^{\mathcal{D}}$ are isomorphic.

If (G, B, T) is quasi-splittable, for example if for G one takes the group $G'_{\text{q-pin.}}$, with its canonical Killing pair, it follows that the image of the class of P is nothing other than the obstruction to the quasi-splitting of G' defined in 3.9.

3.16. — Symmetries

3.16.1. — Let S be a scheme, G a reductive S -group, and $B \supset T$ a Killing pair of G . Put $\mathcal{D} = \underline{\text{Dyn}}(G)$. Recall (Exp. XXII, 5.9.1) that there exists a unique Borel subgroup B^- of G such that $B \cap B^- = T$. If $X \in \Gamma(\mathcal{D}, \mathfrak{g}^{\mathcal{D}})^{\times}$ defines a quasi-pinning of G relative to (B, T) (3.9), then $Y = -X^{-1} \in \Gamma(\mathcal{D}, (\mathfrak{g}^{\mathcal{D}})^{\vee})^{\times}$ defines a quasi-pinning of G relative to (B^-, T) ; one says that it is the *opposite quasi-pinning*.

If $\mathcal{R}$ is a reduced pinned root datum and $\mathcal{E}$ an $\mathcal{R}$ -pinning of the reductive S -group G , one defines an $\mathcal{R}$ -pinning $\mathcal{E}^-$ said to be *opposite to $\mathcal{E}$* as follows: one keeps the same maximal torus T , takes the opposite of the isomorphism $D_S(M) \xrightarrow{\sim} T$, and “pins” by $Y_{\alpha} = -X_{\alpha}^{-1} \in \Gamma(S, \mathfrak{g}^{-\alpha})^{\times}$, for $\alpha \in \Delta$. The quasi-pinning underlying $\mathcal{E}^-$ is the quasi-pinning opposite to the quasi-pinning underlying $\mathcal{E}$.

Remark. In the notation of Exp. XIX, 3.1, if one puts

$$w_\alpha(X_\alpha) = \exp(X_\alpha) \exp(-X_\alpha^{-1}) \exp(X_\alpha),$$

one has $w_\alpha(X_\alpha) = w_{-\alpha}(Y_\alpha)$ (loc. cit., 3.1 (vi)).

Proposition 3.16.2. *Let S be a scheme and G a reductive S -group.*

(i) *Let T be a maximal torus of G ; there exists a unique*

$$i_T \in (\underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}})(S) \subset \text{Aut}_{S\text{-gr.}}(T)$$

such that $i_T(t) = t^{-1}$ for every section t of T .

(ii) *Let (B, T) be a Killing pair of G ; there exists a unique section*

$$w_{B,T} \in (\underline{\text{Norm}}_G(T)/T)(S) = W_G(T)(S)$$

such that $\text{int}(w_{B,T})(B) = B^-$ (with the evident abuse of language).

(iii) *Let $\mathcal{Q} = (B, T, X)$ be a quasi-pinning of G , and $\mathcal{Q}^- = (B^-, T, Y)$ the opposite quasi-pinning; there exists a unique inner automorphism of G*

$$n_{\mathcal{Q}} \in \underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}})(S) \subset \text{Aut}_{S\text{-gr.}}(G)$$

such that $n_{\mathcal{Q}}(\mathcal{Q}) = \mathcal{Q}^-$, i.e. $n_{\mathcal{Q}}(T) = T$, $n_{\mathcal{Q}}(B) = B^-$, $n_{\mathcal{Q}}(X) = Y$.

(iv) *Let $(\mathcal{R}, \mathcal{E})$ be a pinning of G , and $(\mathcal{R}, \mathcal{E}^-)$ the opposite pinning; there exists a unique automorphism of G*

$$u_{\mathcal{E}} \in \text{Aut}_{S\text{-gr.}}(G, T) \subset \text{Aut}_{S\text{-gr.}}(G)$$

such that $u_{\mathcal{E}}(\mathcal{E}) = \mathcal{E}^-$, i.e. $u_{\mathcal{E}}(t) = t^{-1}$ for every section t of T , and $\text{Ad}(u_{\mathcal{E}})X_\alpha = Y_\alpha$ for every $\alpha \in \Delta$.

PROOF. (ii) follows from Exp. XXII, 5.5.5 (ii), then (iii) follows from 3.10, and (iv) follows from Exp. XXIII, 4.1. Finally, to prove (i), one may suppose G pinned. Existence follows from (iv), for example, and uniqueness from the fact that an automorphism of G inducing the identity on T is given by a section of T^{ad} (2.11). $\square$

Corollary 3.16.3. *One has*

$$i_T^2 = w_{B,T}^2 = n_{\mathcal{Q}}^2 = u_{\mathcal{E}}^2 = e.$$

Moreover, i_T (resp. $u_{\mathcal{E}}$) is $\neq e$ if $G \neq e$, and $w_{B,T}$ (resp. $n_{\mathcal{Q}}$) is $\neq e$ if G is not a torus.

Corollary 3.16.4. *In the situation of (iii) (resp. (iv)), $n_{\mathcal{Q}}$ projects to $w_{B,T}$ (resp. $u_{\mathcal{E}}$ projects to i_T) under the canonical morphism*

$$\underline{\text{Norm}}_{\text{ad}(G)}(T^{\text{ad}}) \longrightarrow W_{\text{ad}(G)}(T^{\text{ad}}) \simeq W_G(T),$$

resp.

$$\underline{\text{Aut}}_{S\text{-gr.}}(G, T) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}}.$$

Corollary 3.16.5. *The preceding definitions are compatible with extension of the base, and are functorial under isomorphism (in an evident sense).*

Proposition 3.16.6.

(i) *One can define uniquely for every reductive group G over a scheme S an element*

$$s_G \in \underline{\text{Autext}}(G)(S)$$

in such a way that this construction is functorial in G under isomorphism, is compatible with base change, and whenever T is a maximal torus of the reductive S -group G , s_G is the image of i_T under the canonical morphism

$$\underline{\text{Aut}}_{S\text{-gr.}}(G, T)/T^{\text{ad}} \longrightarrow \underline{\text{Autext}}(G).$$

(ii) *One has $s_G^2 = e$, and s_G is a central element of $\underline{\text{Autext}}(G)$.*

- (iii) Under the conditions of 3.16.2 (ii), if one identifies $\underline{\text{Aut}}_{S\text{-gr.}}(G, B, T)/T^{\text{ad}}$ with $\underline{\text{Autext}}(G)$ (2.2), one has

$$w_{B,T}i_T = i_T w_{B,T} = s_G.$$

- (iv) Under the conditions of 3.16.2 (iv), if one identifies $\text{Aut}_{S\text{-gr.-pin}}(G, \mathcal{R}, \mathcal{E})$ with $\underline{\text{Autext}}(G)$ (1.3 (iii)), one has

$$n_{\mathcal{E}}u_{\mathcal{E}} = u_{\mathcal{E}}n_{\mathcal{E}} = s_G.$$

PROOF. (i) is proved without difficulty by descent. Moreover, since i_T is evidently a central section of square e in $\text{Aut}_{S\text{-gr.}}(T)$, (ii) follows immediately; (iii) is a consequence of (iv) by descent. Finally, under the conditions of (iv), it is clear that $n_{\mathcal{E}}u_{\mathcal{E}} = u_{\mathcal{E}}n_{\mathcal{E}}$ and that this automorphism of G respects the pinning; under the identification made, it is therefore equal to its image in $\underline{\text{Autext}}(G)$; but $n_{\mathcal{E}}$ is inner and $u_{\mathcal{E}}$ projects to s_G . $\square$

Remark 3.16.7. (i) One determines s_G explicitly in each case of the classification by means of (iii): for each irreducible pinned root datum, it suffices to compose the symmetry about the origin with the symmetry in the Weyl group (i.e. the element w_0 of the Weyl group such that $w_0(\Delta) = -\Delta$). One finds the following results: $s_G = e$ except for A_n ($n \geq 2$), D_n (n odd) and E_6 , in which case s_G is the unique nontrivial “outer automorphism”.

(ii) The automorphism $u_{\mathcal{E}}$ is the one used to construct “the compact real forms” in the theory of semisimple Lie algebras.

Remark 3.16.8. In 3.16.1 one has defined an involution on the S -scheme

$$Q = \underline{\text{Isomint}}(G_{\text{q-pin.}}, G)$$

of quasi-pinnings of G (cf. 3.12.1); by transport of structure from $G_{\text{q-pin.}}$ to G , one sees at once that this involution is given by the action of an element of $\text{ad}(G_{\text{q-pin.}})(S)$: the element n_0 defined (3.16.2 (iii)) by the canonical quasi-pinning of $G_{\text{q-pin.}}$.

In the same way, one has defined an involution on the S -scheme

$$P = \underline{\text{Isom}}_{S\text{-gr.}}(\acute{\text{E}}\text{p}_S(\mathcal{R}), G)$$

of $\mathcal{R}$ -pinnings of G (cf. 1.20). Reasoning as before, one sees that this involution is given by the action of the automorphism u_0 of $\acute{\text{E}}\text{p}_S(\mathcal{R})$ defined (3.16.2 (iv)) by the canonical pinning of $\acute{\text{E}}\text{p}_S(\mathcal{R})$.

4. Isotriviality of reductive groups and principal bundles under reductive groups

4.1. — Definitions. Isotriviality theorem

Definition 4.1.1. Let S be a scheme, G an S -group scheme, and P a principal homogeneous bundle under G . One says that P is *locally isotrivial* (resp. *semi-locally isotrivial*) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open subset) there exist an open subset U of S containing s (resp. F) and a finite surjective étale morphism $S' \rightarrow U$ such that $P_{S'}$ is trivial.

Definition 4.1.2. Let S be a scheme and G a reductive S -group. One says that G is *locally isotrivial* (resp. *semi-locally isotrivial*) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open subset) there exist an open subset U of S containing s (resp. F) and a finite surjective étale morphism $S' \rightarrow U$ such that $G_{S'}$ is splittable.

Remark 4.1.3. (i) The equivalence of categories of 1.17 respects, by definition, local (resp. semi-local) isotriviality.

(ii) Add to the conditions of 4.1.1: G locally of finite presentation over S . Then the principal bundle P (or the reductive group G) is locally isotrivial (resp. semi-locally isotrivial)

if and only if for every $S' \rightarrow S$, with S' local (resp. semi-local), $P_{S'}$ is isotrivial (or $G_{S'}$ is isotrivial), i.e. if there exists a finite surjective étale morphism $S'' \rightarrow S'$ such that $P_{S''}$ is trivial (or $G_{S''}$ split).

(iii) In the case of tori, Definition 4.1.2 coincides with that of Exp. IX, 1.1.

4.1.4. — Recall (Exp. XXII, 4.3 and 6.2) that for every reductive group G we have introduced the groups $\text{rad}(G)$, $\text{corad}(G)$ and $\text{der}(G)$. The groups $\text{rad}(G)$ and $\text{corad}(G)$ are tori, which are split when G is split; moreover, there exists an isogeny $\text{rad}(G) \rightarrow \text{corad}(G)$. The S -group $\text{der}(G)$ is semisimple, and one has $G/\text{der}(G) = \text{corad}(G)$; it follows that for every principal homogeneous bundle P under G , $P/\text{der}(G)$ is a principal homogeneous bundle under $\text{corad}(G)$. This being said, one has:

Theorem 4.1.5. *Let S be a scheme and G a reductive S -group of constant type.*

- (i) *The following conditions are equivalent:*
 - (a) *G is locally (resp. semi-locally) isotrivial.*
 - (b) *The torus $\text{rad}(G)$ is so.*
 - (b') *The torus $\text{corad}(G)$ is so.*
 - (c) *The Galois covering of S associated to G (3.11) is so.*
- If T is a maximal torus of G , these conditions are also equivalent to*
 - (d) *The torus T is locally (resp. semi-locally) isotrivial.*
- (ii) *Let P be a principal homogeneous bundle under G ; for P to be locally (resp. semi-locally) isotrivial, it is necessary and sufficient that the principal $\text{corad}(G)$ -bundle $P/\text{der}(G)$ be so.*

Corollary 4.1.6. *The conditions of (i) hold when G is semisimple or when S is locally noetherian and normal (or, more generally, geometrically unibranch). The conditions of (ii) hold when G is locally (resp. semi-locally) isotrivial.*

PROOF. For (i), the first assertion is trivial from (b), and the second follows from (c) and Exp. X, 5.14 and 5.15. For (ii), it suffices to observe that, by theorem 90, a principal bundle under a split torus is semi-locally isotrivial. $\square$

4.2. — Proof: the semisimple case Let us first prove, for later reference:

Proposition 4.2.1. *Let S be a scheme, G a reductive S -group, T a maximal torus of G (resp. B a Borel subgroup, resp. $B \supset T$ a Killing pair of G), P a principal homogeneous bundle under G , and G' the twisted form of G associated to P (via the morphism $\text{int} : G \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G)$). One has canonical isomorphisms*

$$P/\underline{\text{Norm}}_G(T) \xrightarrow{\sim} \underline{\text{Tor}}(G'), \quad P/B \xrightarrow{\sim} \underline{\text{Bor}}(G'), \quad P/T \xrightarrow{\sim} \underline{\text{Kil}}(G').$$

PROOF. By construction, G' is the quotient of $P \times_S G$ by a certain action of G ($(p, g')g = (pg, g^{-1}g'g)$); one therefore has a morphism $P \times_S G \rightarrow G'$, i.e. a morphism

$$P \rightarrow \underline{\text{Hom}}_S(G, G'),$$

which, as one sees at once, factors through a morphism

$$f : P \rightarrow \underline{\text{Isom}}_{S\text{-gr.}}(G, G'),$$

(to verify this assertion, one may suppose $G = P$, in which case one has $G' = G$, $f = \text{int}$). The set-theoretic map $p \mapsto f(p)(T)$ defines a morphism

$$P \rightarrow \underline{\text{Tor}}(G').$$

To verify that this morphism induces an isomorphism $P/\underline{\text{Norm}}_G(T) \xrightarrow{\sim} \underline{\text{Tor}}(G')$ as asserted, one may again suppose $P = G$, in which case one is reduced to Exp. XXII, 5.8.3 (iii). (In fact, loc. cit. ought to be replaced by the above statement.) One reasons likewise for $\underline{\text{Bor}}$ and $\underline{\text{Kil}}$. $\square$

Proposition 4.2.2. *Let S be a semi-local scheme and G a semisimple S -group of constant type.*

- (i) G is isotrivial.
- (ii) Every principal bundle under G is isotrivial.

Let us prove (i). After a finite surjective étale extension of the base, one may, by 3.9.2, suppose G quasi-split. But then G is isotrivial by construction (3.11, the group E is finite). To prove (ii), one may, by (i), suppose G split; one is then reduced to:

Lemma 4.2.3. *Let S be a semi-local scheme. Every principal bundle under a split reductive group is isotrivial.*

PROOF. Indeed, with the notation of 4.2.1, where $B \supset T$ denotes the canonical Killing pair of the split G , the S -scheme $\underline{\text{Kil}}(G')$ has a section, after a finite surjective étale extension of the base, by 3.9.2. One can therefore reduce the structure group of G to T ; now T is split, hence $H^1(S, T) = 0$ (theorem 90). $\square$

Corollary 4.2.4. *Let S be a scheme and*

$$1 \longrightarrow G \longrightarrow G' \longrightarrow G'' \longrightarrow 1$$

an exact sequence of S -group schemes (for (fpqc)), G being semisimple of constant type. Let P be a principal homogeneous bundle under G' , and suppose the associated sheaf⁽²⁴⁾ P/G representable (for example G'' affine over S). For P to be locally isotrivial (resp. semi-locally isotrivial), it is necessary and sufficient that P/G be so (as a bundle under $G''^{(24)}$).

PROOF. If P is trivial, P/G is also trivial, which shows that the condition is necessary. Conversely, suppose S local (resp. semi-local) and P/G isotrivial, hence trivialized by an étale finite surjective extension S' of S . Extending the base to S' , one can reduce the structure group of $P_{S'}$ to $G_{S'}$. But S' is still semi-local and $G_{S'}$ semisimple of constant type, so the bundle obtained is isotrivial (4.2.2). $\square$

4.3. — Proof: general case. Observe first that 4.1.5 (ii) follows immediately by applying 4.2.4 to the exact sequence

$$1 \longrightarrow \text{der}(G) \longrightarrow G \longrightarrow \text{corad}(G) \longrightarrow 1.$$

Let us therefore prove (i). If G is split, $\text{rad}(G)$ and $\text{corad}(G)$ are split, as is $\text{rev}(G)$; thus (a) implies (b), (b'), and (c).

4.3.1. — One has (c) $\Rightarrow$ (a). Let $\mathcal{R}$ be the type of G ; one has an exact sequence

$$1 \longrightarrow \text{ad}(\dot{\text{Ep}}_S(\mathcal{R})) \longrightarrow A_S(\mathcal{R}) \longrightarrow E_S \longrightarrow 1.$$

Applying 4.2.4 to the bundle $P = \underline{\text{Isom}}_{S\text{-gr}}(\dot{\text{Ep}}_S(\mathcal{R}), G)$ and the associated bundle $\text{rev}(G) = P/\text{ad}(\dot{\text{Ep}}_S(\mathcal{R}))$, one has (c) $\Rightarrow$ (a).

4.3.2. — One has (b) $\Rightarrow$ (a). It suffices for us to prove that if $\text{rad}(G)$ is split, G is semi-locally isotrivial. Now let $G_0 = \dot{\text{Ep}}_S(\mathcal{R})$; consider the category of pairs (G', f) , where G' is a form of G_0 and f an isomorphism of $\text{rad}(G_0)$ onto $\text{rad}(G)$.

One knows (2.16) that this category is equivalent to the category of principal homogeneous bundles under a certain S -group H , an extension of a finite twisted constant group by a semisimple group. It suffices for us to prove that every principal bundle under H is semi-locally isotrivial; now this follows at once from 4.2.4.

4.3.3. — One has (b') $\Rightarrow$ (a). One can reason as before (in fact the same group H will occur). One can also see that (b) and (b') are equivalent: a torus isogenous to a locally split torus is also locally split (cf. Exp. IX, 2.11 (iii)).

⁽²⁴⁾**N.D.E.** We have replaced “bundle” by “sheaf”, and then G' by G'' .

4.3.4. — One has (d) $\Rightarrow$ (a). It suffices to prove that a group of constant type possessing a split maximal torus is semi-locally isotrivial; now this follows at once from 2.14.

4.3.5. — One has (a) $\Rightarrow$ (d). It suffices to prove that a maximal torus of a split group is semi-locally isotrivial. More precisely, one has:

Lemma 4.3.6. *Let S be a semi-local scheme, G a reductive S -group, T_0 a split maximal torus of G , $W_0 = \underline{\text{Norm}}_G(T_0)/T_0$ (it is a locally constant S -group, and constant if G is of constant type, by 2.14), and T a maximal torus of G .*

There exists a morphism $S' \rightarrow S$ which is principal homogeneous under W_0 (hence étale, finite and surjective, and even principal Galois if G is of constant type), such that $T_{S'}$ is conjugate to $(T_0)_{S'}$ by an element of $G(S')$ (and therefore in particular split).

PROOF. Indeed, one knows that $\underline{\text{Transp}}_G(T_0, T)$ is a principal homogeneous bundle under $\underline{\text{Norm}}_G(T_0)$ (cf. for example Exp. XI, 5.4 bis). Put $S' = \underline{\text{Transp}}_G(T_0, T)/T_0$; it is a principal homogeneous bundle under W_0 . Extending the base from S to S' , one can reduce the structure group of $\underline{\text{Transp}}_G(T_0, T)$ to T_0 . Now S' is semi-local and T_0 split, so $\underline{\text{Transp}}_G(T_0, T)$ has a section over S' .⁽²⁵⁾ $\square$

4.4. — Use of the existence of maximal tori By using Grothendieck's theorem on the existence of maximal tori (Exp. XIV, 3.20), one can considerably sharpen the preceding results. Let us state at once:

Theorem 4.4.1. *Let S be a semi-local scheme, $\mathcal{R}$ a reduced pinned root datum, W its Weyl group, and E the group of its automorphisms. (Recall that E acts naturally on W and that the semidirect product $A = W \cdot E$ identifies with the group of automorphisms of the unpinned $\mathcal{R}$, cf. Exp. XXI, 6.7.2.)*

- (i) *Every principal homogeneous bundle under $\dot{\text{Ep}}_S(\mathcal{R})$ is trivialized by a principal Galois covering $S' \rightarrow S$ with group W .*
- (ii) *Let G be a reductive S -group of type $\mathcal{R}$; let $\text{rev}(G) = \underline{\text{Isomext}}(\dot{\text{Ep}}_S(\mathcal{R}), G)$ be the associated Galois covering of S with group E . Let W_0 be the form of W_S associated to $\text{rev}(G)$. There exists a morphism $S' \rightarrow S$, which is a principal homogeneous bundle under W_0 , such that $G_{S'}$ is quasi-split (i.e. possesses a Borel subgroup).*
- (iii) *Every reductive S -group of type $\mathcal{R}$ is split by a principal Galois covering $\tilde{S} \rightarrow S$ with group A .*

Let us first state:

Proposition 4.4.2. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , P a principal homogeneous bundle under G , and G' the twisted form of G associated to P (one then has, cf. 4.2.1, a canonical isomorphism $P/\underline{\text{Norm}}_G(T) \xrightarrow{\sim} \underline{\text{Tor}}(G')$). Let T' be a maximal torus of G' , defining a composite morphism*

$$S \rightarrow \underline{\text{Tor}}(G') \xrightarrow{\sim} P/\underline{\text{Norm}}_G(T).$$

Consider the canonical morphisms

$$P \rightarrow P/T \rightarrow P/\underline{\text{Norm}}_G(T)$$

and take their inverse images under the preceding morphism:

$$\begin{array}{ccccc} P & \longrightarrow & P/T & \longrightarrow & P/\underline{\text{Norm}}_G(T) \\ \uparrow & & \uparrow & & \uparrow \\ H & \longrightarrow & S' & \longrightarrow & S. \end{array}$$

⁽²⁵⁾N.D.E. By “theorem 90”.

Then S' (resp. H) is a principal homogeneous bundle over S (resp. S') under $W_G(T)$ (resp. $T_{S'}$). Moreover, if one lets $W_G(T)$ act on T in the evident way, the bundle associated to S' is isomorphic to T' .

The first two assertions are trivial, and the last is proved as is the corresponding assertion of 2.6.

Corollary 4.4.3. *Let S be a semi-local scheme, G a reductive S -group, and T a maximal torus of G . Suppose that one of the following two conditions holds:*

- (i) T is split.
- (ii) T is contained in a Borel subgroup of G , and G is either adjoint or simply connected.

Let moreover P be a principal homogeneous bundle under G . There exists an S -scheme S' , which is a principal homogeneous bundle under $W_G(T)$, such that $P_{S'}$ is trivial.

PROOF. Indeed, if G' is the form of G associated to P , then G' possesses a maximal torus T' (Exp. XIV, 3.20). Taking up the notation of the preceding proposition, one sees that $H^1(S', T_{S'}) = 0$ (by theorem 90 for (i), and by 3.14 for (ii)). The morphism $H \rightarrow S'$ has a section, so $P_{S'}$ also has a section over S' . $\square$

Let us now prove the theorem. Assertion (i) is a special case of the preceding corollary (take $G = \text{Ép}_S(\mathcal{R})$, equipped with its canonical split torus). Let us prove (ii). One knows (3.12) that G is an inner twisted form of

$$G_0 = G_{\text{rev}(G)/S}^{\text{qÉp}}(\mathcal{R}).$$

If T_0 is the canonical maximal torus of G_0 , $W_{G_0}(T_0)$ is indeed the group W_0 described in the statement. The form G of G_0 corresponds to a principal homogeneous bundle P under $\text{ad}(G_0)$ ($P = \text{Isomint}(G_0, G)$). The group $W_{\text{ad}(G_0)}(T_0^{\text{ad}})$ is canonically isomorphic to W_0 , and one obtains the desired result by applying 4.4.3 to the situation $(\text{ad}(G_0), T_0^{\text{ad}}, P)$, hypothesis (ii) of 4.4.3 being satisfied. Let us finally prove (iii). Take up the notation of (ii); one has a diagram

$$\begin{array}{ccc} & \text{rev}(G) & \\ E_S \downarrow & & \\ S & \xleftarrow{W_0} & S'. \end{array}$$

One knows that $G_{S'}$ is isomorphic to $(G_0)_{S'}$ and that $(G_0)_{\text{rev}(G)}$ is splittable. If one puts $\bar{S} = S' \times_S \text{rev}(G)$, $G_{\bar{S}}$ is indeed splittable, and it only remains to verify that $\bar{S}$ is indeed a principal Galois covering of S with group A , which follows from the more general lemma below (naturally valid in any site):

Lemma 4.4.4. *Let S be a scheme, G and H two S -group schemes, $G \rightarrow \underline{\text{Aut}}_{S\text{-gr.}}(H)$ an action of G on H , E a principal homogeneous G -bundle, and F a principal homogeneous H_0 -bundle, where H_0 is the form of H associated to E . Then $E \times_S F$ is naturally equipped with a structure of principal homogeneous bundle under the semidirect product $H \cdot G$.*

PROOF. Denote by $(e, g) \mapsto eg$ (resp. $(f, u) \mapsto fu$) the (right) action of G on E (resp. of H_0 on F). Denote by

$$p : E \times_S H \rightarrow H_0$$

the canonical projection (H_0 is by definition the quotient of $E \times_S H$ by G acting on it according to the formula $(e, h)g = (eg, g^{-1}hg)$). Consider the morphism

$$r : E \times_S F \times_S H \times_S G \rightarrow E \times_S F$$

defined set-theoretically by

$$r(e, f, h, g) = (eg, fp(e, h)).$$

The morphism r does indeed define an action of the semidirect product $H \cdot G$ on $E \times_S F$. Indeed, one has set-theoretically

$$r(r(e, f, h, g), h', g') = (egg', fp(e, h)p(eg, h'));$$

but

$$p(e, h)p(eg, h') = p(e, h)p(e, gh'g^{-1}) = p(e, hgh'g^{-1}),$$

whence

$$r(r(e, f, h, g), h', g') = r(e, f, hgh'g^{-1}, gg'),$$

which is what had to be proved.

To prove now that this law is indeed a principal homogeneous bundle law, one may suppose that E and F are trivial, in which case one sees at once that $E \times_S F$ is also a trivial bundle. $\square$

4.5. — Isotriviality of maximal tori of semisimple groups⁽²⁶⁾ Let us bring out here the following result, implicitly contained in 2.6 (cf. N.D.E. (13)) and mentioned at the end of Exp. IX 1.2. Recall (XXIII 5.11) that, for every scheme S and every reduced root datum $\mathcal{R}$, one denotes by $\dot{\text{Ep}}_S(\mathcal{R})$ the unique pinned S -group of type $\mathcal{R}$ (which exists by Exp. XXV) and by $T_S(\mathcal{R})$ its canonical maximal torus.

Proposition 4.5.1. *Let G be a semisimple S -group of constant type $\mathcal{R}$; suppose that G possesses a maximal torus T (which is the case if S is semi-local, by XIV 3.20).*

- (i) *Then $P = \underline{\text{Isom}}_{S\text{-gr.}}(\dot{\text{Ep}}_S(\mathcal{R}), T_S(\mathcal{R}); G, T)/T_S(\mathcal{R})^{\text{ad}}$ is a principal S -bundle under the finite group $\text{Aut}(\mathcal{R})$, and T_P is a split maximal torus of G_P . Consequently, T is isotrivial.*
- (ii) *Moreover, for every finite set F of points of P contained in an affine open subset, there exists an open subset U of P containing F such that G_U is split.*

PROOF. The exact sequence

$$1 \longrightarrow T_S(\mathcal{R})^{\text{ad}} \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{Ep}}_S(\mathcal{R}), T_S(\mathcal{R})) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(T_S(\mathcal{R}))$$

induces for every $S' \rightarrow S$ a morphism of bundles $P_{S'} \rightarrow \underline{\text{Isom}}_{S'\text{-gr.}}(T_{S'}(\mathcal{R}), T_{S'})$, so if $P_{S'}$ has a section, then $T_{S'}(\mathcal{R})$ and $T_{S'}$ are isomorphic. This is in particular the case for $S' = P$.

Observe moreover that $\underline{\text{Aut}}_{S\text{-gr.}}(\dot{\text{Ep}}_S(\mathcal{R}), T_S(\mathcal{R}))/T_S(\mathcal{R})^{\text{ad}}$ is nothing other than the constant S -group $\text{Aut}(\mathcal{R})_S$, and that P identifies with the $\text{Aut}(\mathcal{R})_S$ -torsor $\underline{\text{Isom}}(\mathcal{R}, \tilde{\mathcal{R}}(G, T))$, where $\tilde{\mathcal{R}}(G, T)$ denotes the twisted root datum associated to (G, T) (cf. XXII 1.10).

Then (ii) follows from (i) and from 2.14, taking account of the fact that the $\mathfrak{g}^\alpha$ (α ranging over the roots of $\mathcal{R}$) are free $\mathcal{O}_U$ -modules of rank 1 on a sufficiently small affine open subset U containing F . $\square$

5. Canonical decomposition of an adjoint or simply connected group

In this section we shall use the results of no. 1 to generalize to schemes a classical decomposition of adjoint (resp. simply connected) groups. To avoid overburdening the exposition indefinitely, the proofs are sketched and the details left to the reader; in fact these are always entirely standard proofs from the theory of principal bundles: reduction of the structure group, twisting, ...

5.1. — Recall (Exp. XXI, 7.4) that a Dynkin diagram is the disjoint union of its connected components, which are Dynkin diagrams. Moreover, every nonempty connected Dynkin diagram corresponding to a root datum is isomorphic to one of the standard diagrams $(A_n, B_n, \dots, G_2)$ exhibited in Exp. XXI, 7.4.6. In what follows we shall only be concerned with Dynkin diagrams whose connected components are of one of the preceding types. Let $\mathbf{T}$ be

⁽²⁶⁾N.D.E. We have added this no. 4.5.

the set of these standard diagrams. For every Dynkin diagram D , let $n(t) = n_D(t)$ be the number of connected components of D isomorphic to t , where $t \in \mathbf{T}$. The *type* of D is by definition $\sum_{t \in \mathbf{T}} n_D(t)t$.

A Dynkin diagram of type t is said to be *simple of type t* , and a Dynkin diagram of type nt is said to be *isotypic of type t* . Let D_0 be the set of connected components of D and let

$$a : D_0 \longrightarrow \mathbf{T}$$

be the evident map. The type of D is nothing other than $\sum_{x \in D_0} a(x)$.

5.2. — Let S be a scheme and D a *Dynkin S -scheme* (satisfying the restrictive condition stated above). The cokernel of the pair of morphisms $L \rightrightarrows D$ ($L =^{(27)}$ scheme of bonds of D) is denoted by D_0 . It is the “scheme of connected components” of D (it exists trivially when D is constant; the general case follows by descent); it is a finite twisted constant S -scheme. One has a canonical morphism

$$a : D_0 \longrightarrow \mathbf{T}_S.$$

For $t \in \mathbf{T}$, put $a^{-1}(t) = D_{0,t}$; it is a subscheme of D_0 , whose inverse image in D , denoted by D_t , is the *isotypic component of type t* of the Dynkin scheme D . Each D_t is a subscheme of D , and one has

$$D = \coprod_{t \in \mathbf{T}} D_t.$$

Observe that the degree of $D_{0,t}$ at $s \in S$ is $n(s, t)$, if the type of D at s is $\sum_t n(s, t)t$.

5.3. — In what follows we shall consider only adjoint (resp. simply connected) semisimple groups. To simplify the language, we shall state the results for adjoint groups; *all the statements will remain valid* if one everywhere substitutes *simply connected* for *adjoint*.

Recall that a reduced adjoint root datum is determined up to isomorphism by the type of its Dynkin diagrams. One will therefore say that an adjoint root datum $\mathcal{R}$ (resp. an adjoint semisimple group G) is of *type $\sum n(t)t$* if its Dynkin diagrams are so (resp. if its type is given by an adjoint root datum of type $\sum n(t)t$). One will say that $\mathcal{R}$ or G is *simple of type t* (resp. *isotypic of type t*) if its type is t (resp. nt , $n > 0$).

If G is an adjoint semisimple group, we shall use the symbols $\underline{\text{Dyn}}_0(G)$ and $\underline{\text{Dyn}}_{0,t}(G)$ in the sense defined in 5.2.

5.4. — Let t_i , $i = 1, 2, \dots, n$, be distinct elements of $\mathbf{T}$, and let $\mathcal{R}_i$ be a pinned adjoint root datum isotypic of type t_i . Consider the product $\mathcal{R} = \mathcal{R}_1 \times \dots \times \mathcal{R}_n$ (Exp. XXI, 6.4.1). Let S be a scheme such that the various $\acute{\text{E}}\text{p}_S(\mathcal{R}_i)$ exist (cf. Exp. XXV). One verifies at once that there exists a canonical isomorphism

$$(*) \quad \acute{\text{E}}\text{p}_S(\mathcal{R}) = \acute{\text{E}}\text{p}_S(\mathcal{R}_1) \times_S \dots \times_S \acute{\text{E}}\text{p}_S(\mathcal{R}_n).$$

Moreover, if E_i denotes the group of automorphisms of $\mathcal{R}_i$, E the group of automorphisms of $\mathcal{R}$, and D_i (resp. D) the Dynkin diagram of $\mathcal{R}_i$ (resp. $\mathcal{R}$), one has isomorphisms:

$$E_i \simeq \text{Aut}(D_i), \quad D = \coprod D_i, \quad E \simeq \prod E_i \simeq \text{Aut}(D).$$

Combining with $(*)$ and 1.4, one sees that $(*)$ induces an isomorphism

$$A_S(\mathcal{R}) \simeq A_S(\mathcal{R}_1) \times_S \dots \times_S A_S(\mathcal{R}_n).$$

Proposition 5.5. *Let S be a scheme and G an adjoint semisimple S -group. There exists a unique decomposition*

$$G \simeq \prod_{t \in \mathbf{T}} G_t,$$

⁽²⁷⁾**N.D.E.** We have replaced R by L , as in 3.2.

where G_t is an adjoint semisimple S -group isotypic of type t . Moreover, the preceding decomposition induces isomorphisms

$$\underline{\mathrm{Dyn}}_t(G) \simeq \underline{\mathrm{Dyn}}(G_t), \quad \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \simeq \prod_{t \in \mathbf{T}} \underline{\mathrm{Aut}}_{S\text{-gr.}}(G_t).$$

PROOF. This has indeed been proved above when G is split. In the general case, one may suppose G of constant type $\mathcal{R}$. Using the preceding decomposition of $A_S(\mathcal{R})$ and 1.17, one deduces the desired decomposition of G . The other results are then proved by descent. $\square$

Remark 5.6. More generally, if G and H are two adjoint semisimple groups, one has canonical isomorphisms as follows (the diagram is commutative):

$$\begin{array}{ccc} \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H) & \xrightarrow{\sim} & \prod_{t \in \mathbf{T}} \underline{\mathrm{Isom}}_{S\text{-gr.}}(G_t, H_t) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Isomext}}(G, H) & \xrightarrow{\sim} & \prod_{t \in \mathbf{T}} \underline{\mathrm{Isomext}}(G_t, H_t) \\ \downarrow \wr & & \downarrow \wr \\ \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\underline{\mathrm{Dyn}}(G), \underline{\mathrm{Dyn}}(H)) & \xrightarrow{\sim} & \prod_{t \in \mathbf{T}} \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\underline{\mathrm{Dyn}}(G_t), \underline{\mathrm{Dyn}}(H_t)). \end{array}$$

Remark 5.7. One can give an intrinsic characterization of G_t , which we state below without proof: it is the largest normal (and indeed characteristic) reductive subgroup of G isotypic of type t .

The preceding proposition allows one to reduce the study of adjoint semisimple groups to that of isotypic adjoint semisimple groups. It is this study that we shall carry out below.

5.8. — Let $\mathcal{R}$ be a *simple* reduced pinned adjoint root datum of type t , I a nonempty finite set, and $\mathcal{R}^I$ the product root datum of copies $\mathcal{R}_i$ of $\mathcal{R}$, for $i \in I$. Denote by E the group of automorphisms of $\mathcal{R}$, which identifies with the group of automorphisms of the Dynkin diagram D of $\mathcal{R}$. The Dynkin diagram of $\mathcal{R}^I$ identifies with D^I , which shows that one has an exact sequence:

$$1 \longrightarrow \mathrm{Aut}(D)^I \longrightarrow \mathrm{Aut}(D^I) \longrightarrow \mathrm{Aut}(I) \longrightarrow 1,$$

where $\mathrm{Aut}(I)$ denotes the group of permutations of I . It follows that the canonical isomorphism

$$\dot{\mathrm{Ep}}_S(\mathcal{R})^I \simeq \dot{\mathrm{Ep}}_S(\mathcal{R}^I)$$

induces an exact sequence

$$1 \longrightarrow A_S(\mathcal{R})^I \longrightarrow A_S(\mathcal{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S) \longrightarrow 1,$$

the last group being the constant S -group associated to $\mathrm{Aut}(I)$. Observe moreover that I identifies canonically with the set of connected components of D^I . If G is a semisimple S -group of type $\mathcal{R}^I$, defining (cf. 1.17) a principal homogeneous bundle P under $A_S(\mathcal{R}^I)$, the definition of $\underline{\mathrm{Dyn}}(G)$ by descent (3.7), and that of $\underline{\mathrm{Dyn}}_0(G)$ (5.2), show that the bundle associated to P by the morphism $A_S(\mathcal{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S)$ is nothing other than $\underline{\mathrm{Isom}}_S(I_S, \underline{\mathrm{Dyn}}_0(G))$, corresponding to the form $\underline{\mathrm{Dyn}}_0(G)$ of I_S . Using again the equivalence of categories 1.17 and the preceding exact sequence, one deduces by a formal argument that there exist a reductive $\underline{\mathrm{Dyn}}_0(G)$ -group of type $\mathcal{R}$, say G_0 , and an S -isomorphism $G \simeq \prod_{\underline{\mathrm{Dyn}}_0(G)/S} G_0$.

Proposition 5.9. *Let S be a scheme and G an isotypic adjoint semisimple S -group of type t . There exist a simple adjoint semisimple $\underline{\mathrm{Dyn}}_0(G)$ -group G_0 of type t , and an S -isomorphism (unique)*

$$G \simeq \prod_{\underline{\mathrm{Dyn}}_0(G)/S} G_0.$$

Moreover, this isomorphism induces an exact sequence

$$1 \longrightarrow \prod_{\underline{\text{Dyn}}_0(G)/S} \underline{\text{Aut}}_{S\text{-gr.}}(G_0) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Dyn}}_0(G)) \longrightarrow 1.$$

PROOF. One may evidently suppose G of constant type nt . The first assertion has already been proved (the uniqueness assertion is evident). The second then follows from the split case by descent. $\square$

One can combine 5.5 and 5.9:

Proposition 5.10. *Let S be a scheme, G an adjoint semisimple S -group, and $\mathcal{D} = \underline{\text{Dyn}}(G)$ its Dynkin scheme.*

(i) *There exists a canonical decomposition*

$$G \simeq \prod_{t \in \mathbf{T}} \prod_{\mathcal{D}_{0,t}/S} G_{0,t} \simeq \prod_{\underline{\text{Dyn}}_0(G)/S} G_0,$$

where each $G_{0,t}$ is a simple adjoint $\underline{\text{Dyn}}_{0,t}(G)$ -group of type t (resp. where G_0 is an adjoint semisimple $\underline{\text{Dyn}}_0(G)$ -group whose type at $x \in \underline{\text{Dyn}}_0(G)$ is $a(x) \in \mathbf{T}$ (the morphism $a : \underline{\text{Dyn}}_0(G) \rightarrow \mathbf{T}_S$ was defined in 5.2)).

(ii) *The preceding decompositions induce isomorphic exact sequences (written vertically), where $\underline{\text{Aut}}_{S,a}(\mathcal{D}_0)$ denotes the scheme of automorphisms of $\underline{\text{Dyn}}_0(G)$ which commute with the morphism $a : \underline{\text{Dyn}}_0(G) \rightarrow \mathbf{T}_S$:*

$$\begin{array}{ccc} 1 & & 1 \\ \downarrow & & \downarrow \\ \prod_{t \in \mathbf{T}} \prod_{\mathcal{D}_{0,t}/S} \underline{\text{Aut}}_{\text{gr.}}(G_{0,t}) & \xrightarrow{\sim} & \prod_{\underline{\text{Dyn}}_0(G)/S} \underline{\text{Aut}}_{\text{gr.}}(G_0) \\ \downarrow & & \downarrow \\ \underline{\text{Aut}}_{S\text{-gr.}}(G) & \longrightarrow & \underline{\text{Aut}}_{S\text{-gr.}}(G) \\ \downarrow & & \downarrow \\ \prod_{t \in \mathbf{T}} \underline{\text{Aut}}_S(\underline{\text{Dyn}}_{0,t}(G)) & \xrightarrow{\sim} & \underline{\text{Aut}}_{S,a}(\underline{\text{Dyn}}_0(G)) \\ \downarrow & & \downarrow \\ 1 & & 1 \end{array}$$

Corollary 5.11. *Under the preceding conditions, the following three categories are equivalent:*

- (i) *the category of principal homogeneous bundles under G .*
- (ii) *the category of principal homogeneous bundles under G_0 .*
- (iii) *the product category, for $t \in \mathbf{T}$, of the categories of principal homogeneous bundles under the $G_{0,t}$.*

This follows formally from the preceding decompositions and from 8.4.

Corollary 5.12. *One has canonical isomorphisms*

$$\underline{\text{Tor}}(G) \simeq \prod_{t \in \mathbf{T}} \prod_{\mathcal{D}_{0,t}/S} \underline{\text{Tor}}(G_{0,t}) \simeq \prod_{\underline{\text{Dyn}}_0(G)/S} \underline{\text{Tor}}(G_0),$$

and likewise on replacing $\underline{\text{Tor}}$ by $\underline{\text{Bor}}$ (resp. $\underline{\text{Kil}}$).

Trivial from the split case.

Remark 5.13. The canonical morphism

$$\underline{\text{Dyn}}(G) \longrightarrow \underline{\text{Dyn}}_0(G)$$

allows one to consider $\underline{\text{Dyn}}(G)$ as a Dynkin $\underline{\text{Dyn}}_0(G)$ -scheme; in fact it is the Dynkin scheme $\underline{\text{Dyn}}(G_0)$ of G_0 .

Likewise, if $T \subset B$ is a Killing pair of G , canonically corresponding to the Killing pair $T_0 \subset B_0$ of G_0 , one verifies that the obstructions to quasi-splitting G and G_0 , which lie (3.9) in $\text{Pic}(\underline{\text{Dyn}}(G)) = \text{Pic}(\underline{\text{Dyn}}(G_0))$, coincide. One deduces:

Corollary 5.14. *The following conditions are equivalent:*

- (i) G is quasi-splittable,
- (ii) G_0 is quasi-splittable,
- (iii) each $G_{0,t}$, $t \in \mathbf{T}$, is quasi-splittable.

6. Automorphisms of Borel subgroups of reductive groups

Lemma 6.1. *Let S be a scheme, (G, T, M, R) a split S -group, Δ a system of simple roots of R , and B the corresponding Borel subgroup. Then B^u is generated as an (fppf) sheaf of groups by the U_α , $\alpha \in \Delta$.*

Let H be the subsheaf of groups of B^u generated by the U_α , $\alpha \in \Delta$. Let us prove $H \supset U_\beta$ ($\beta \in R_+$) by induction on the integer $\text{ord}(\beta) = \text{ord}_\Delta(\beta) > 0$ (cf. Exp. XXI, 3.2.15). The assertion holds by hypothesis for $\text{ord}(\beta) = 1$. Suppose therefore $\text{ord}(\beta) > 1$ and $U_\gamma \subset H$ whenever $\text{ord}(\gamma) < \text{ord}(\beta)$. There exists $\alpha \in \Delta$ such that $\beta - \alpha \in R_+$ (Exp. XXI, 3.1.1). Let p be the largest integer such that $\beta - p\alpha = \beta' \in R_+$. One has $U_\alpha, U_{\beta'} \subset H$, $\beta' - \alpha \notin R$. One is therefore reduced to proving:

Lemma 6.2. *Take up the notation of Exp. XXIII, 6.4. Suppose $p = 1$, i.e. $\beta - \alpha$ is not a root. If H is a subsheaf of groups of B^u such that $U_\alpha, U_\beta \subset H$, then $U_{i\alpha+j\beta} \subset H$ whenever $i\alpha + j\beta \in R$, $i > 0$, $j > 0$.*

PROOF. Let us distinguish four cases according to the value of $q = 0, 1, 2, 3$. In what follows x and y denote two arbitrary sections of $\mathbf{G}_{a,S'}, S' \rightarrow S$.

If $q = 0$, there is nothing to prove.

If $q = 1$, one has $p_{\alpha+\beta}(\pm x) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S')$, hence $U_{\alpha+\beta} \subset H$.

If $q = 2$, one similarly has

$$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S'),$$

whence, after changing certain signs,

$$F(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y) \in H(S').$$

Since $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ commute, one then has $F(x, 1)F(1, -x) = p_{2\alpha+\beta}(x^2 - x)$.⁽²⁸⁾ If $a \in \mathbf{G}_a(S)$, the equation $X^2 - X = a$ defines a free surjective extension of S (it is $\text{Spec } \mathcal{O}_S[X]/(X^2 - X - a)$); one therefore has $p_{2\alpha+\beta}(a) \in H(S')$, hence $U_{2\alpha+\beta} \subset H$, hence also $U_{\alpha+\beta} \subset H$ (since $F(1, a) \in H(S')$).

If $q = 3$, one similarly has

$$F(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2) \in H(S');$$

and since

$$p_{3\alpha+2\beta}(\pm x) = F(1, 1)^{-1}p_\beta(-x)F(1, 1)p_\beta(x) \in H(S'),$$

one obtains that $U_{3\alpha+2\beta} \subset H$ and therefore

$$K(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y) \in H(S').$$

⁽²⁸⁾**N.D.E.** We have corrected $F(-1, x)$ to $F(1, -x)$.

Then computing

$$K(x+y, 1)K(-x, 1)K(1, y)^{-1}$$

modulo $U_{3\alpha+2\beta}$, one finds

$$p_{2\alpha+\beta}(2x^2 + y^2 + 2xy - y)p_{3\alpha+\beta}(y^3 + 3xy^2 + 3x^2y - y) \in H(S').$$

If $a \in \mathbf{G}_a(S)$, the “equations”

$$\begin{aligned} x^2 &= -xy - y + 1 - a \\ y^2 &= 3y - 2 + 3a \end{aligned}$$

define a free surjective extension of S ; one then has $y^3 + 3xy^2 + 3x^2y - y = 0$ and the preceding expression gives $p_{2\alpha+\beta}(a) \in H(S')$.⁽²⁹⁾ One has therefore proved that H contains $U_{2\alpha+\beta}$ and $U_{3\alpha+2\beta}$ and that

$$E(x, y) = p_{\alpha+\beta}(xy)p_{3\alpha+\beta}(x^3y) \in H(S').$$

Since $p_{\alpha+\beta}(xy)$ and $p_{3\alpha+\beta}$ commute, one is reduced to the preceding computation, i.e. $E(1, x)E(1, -x) = p_{3\alpha+\beta}(x^3 - x)$, whence $U_{3\alpha+\beta} \subset H$, then $U_{\alpha+\beta} \subset H$. $\square$

Remark 6.2.1. The preceding proof shows that one could have replaced the hypothesis “ H contains U_α and U_β ” by “ H contains U_α or U_β , and H is invariant under $\text{int}(U_\alpha)$ and $\text{int}(U_\beta)$ ”.

Theorem 6.3. *Let S be a scheme, G and G' two semisimple S -groups, and B (resp. B') a Borel subgroup of G (resp. G'). Every isomorphism $B \xrightarrow{\sim} B'$ is induced by a unique isomorphism $G \xrightarrow{\sim} G'$.*

PROOF. The assertion is local for the étale topology and one may suppose G split: $G = (G, T, M, R)$, B being defined by the system R_+ of positive roots of R . The given isomorphism $u : B \xrightarrow{\sim} B'$ induces an isomorphism of T onto a maximal torus T' of B' , hence of G' . The given isomorphism $T \simeq D_S(M)$ gives an isomorphism $T' \simeq D_{S'}(M)$ in which the elements of R_+ become the constant roots of B' relative to T' , hence the elements of R the constant roots of G' relative to T' . Since G and G' are semisimple, the coroots are determined by duality (Exp. XXI, 1.2.5), which proves that (T', M, R) is a splitting of G' such that $\mathcal{R}(G) = \mathcal{R}(G')$.

Applying Exp. XXIII, 5.1 (Uniqueness theorem), one deduces that there exists a unique isomorphism $G \xrightarrow{\sim} G'$ agreeing with u on T and the U_α , $\alpha \in \Delta$. By 5.1, the restriction of this isomorphism to B is u . $\square$

Remark 6.3.1. (i) Using Exp. XXII, 4.1.9 and reasoning as in loc. cit. 4.2.12, one can replace “isomorphism” in the statement of the theorem by “isogeny” (resp. “central isogeny”).

(ii) The theorem is false for reductive groups. Take for example $G = G' = \text{SL}_2 \times \mathbf{G}_m$, identified with the group of the following matrices:

$$\left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & h \end{pmatrix} \mid ad - bc = 1, h \text{ invertible} \right\};$$

take for $B = B'$ the Borel subgroup defined by $c = 0$ and for u the following automorphism of B : $u(a, b, d, h) = (a, b, d, ah)$.

Corollary 6.4. *The functor $(G, B) \mapsto B$ from the category consisting of pairs (G, B) , where G is a semisimple S -group and B a Borel subgroup of G , to the category of S -group schemes (the morphisms are the isomorphisms) is fully faithful.*

Corollary 6.5. *Let S be a scheme, G a semisimple S -group, B a Borel subgroup of G , and B' an S -group locally isomorphic to B for (fpqc). Then B' is locally isomorphic to B for the*

⁽²⁹⁾**N.D.E.** We have corrected $p_{3\alpha+\beta}(a)$ to $p_{2\alpha+\beta}(a)$, and given the end of the argument in detail.

local finite étale topology⁽³⁰⁾ and there exists a semisimple S -group G' of which B' is a Borel subgroup; G' is unique up to a unique isomorphism inducing the identity on B' .

Corollary 6.6. *Let S be a scheme, G a semisimple S -group, and B a Borel subgroup of G . Then $\underline{\text{Aut}}_{S\text{-gr.}}(B)$ is representable by a smooth affine S -scheme, $\underline{\text{Autext}}(B)$ is representable by a finite étale S -scheme, and the evident morphisms induce an isomorphism of exact sequences*

$$\begin{array}{ccccccc} 1 & \longrightarrow & B^{\text{ad}} & \longrightarrow & \underline{\text{Aut}}_{S\text{-gr.}}(G, B) & \longrightarrow & \underline{\text{Autext}}(G) \longrightarrow 1 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ 1 & \longrightarrow & B^{\text{ad}} & \longrightarrow & \underline{\text{Aut}}_{S\text{-gr.}}(G) & \longrightarrow & \underline{\text{Autext}}(B) \longrightarrow 1. \end{array}$$

This follows at once from 2.1 and the preceding results. We leave to the reader the task of developing the analogues of the results of nos. 1, 2, 3, 4 in the above setting.

Remark 6.7. If S is a scheme and B an S -group, B can therefore be a Borel subgroup of only one semisimple group, completely determined by B . It therefore remains to characterize the S -groups B which are indeed Borel subgroups of semisimple groups.⁽³¹⁾

7. Representability of the functors $\underline{\text{Hom}}_{S\text{-gr.}}(G, H)$, for reductive G

7.1. — The split case

7.1.1. — Let S be a scheme, G a pinned S -group, T its maximal torus, Δ the set of simple roots, and, for $\alpha \in \Delta$,

$$u_\alpha \in U_\alpha^\times(S), \quad w_\alpha \in \underline{\text{Norm}}_G(T)(S),$$

the elements defined by the pinning. Let, on the other hand, H be an S -group scheme; we are interested in the functor $\underline{\text{Hom}}_{S\text{-gr.}}(G, H)$, and more precisely in the morphism

$$q : \underline{\text{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\text{Hom}}_{S\text{-gr.}}(T, H).$$

Let therefore

$$f_T : T \longrightarrow H$$

be a morphism of S -groups, and consider $q^{-1}(f_T) = \mathcal{F}$. It is the functor defined by

$$\mathcal{F}(S') = \{f \in \underline{\text{Hom}}_{S'\text{-gr.}}(G_{S'}, H_{S'}) \mid f = (f_T)_{S'} \text{ on } T_{S'}\}.$$

One has a morphism of S -functors

$$i : \mathcal{F} \longrightarrow H^{2n},$$

where $n = \text{Card}(\Delta)$, defined set-theoretically by $i(f) = (f(u_\alpha), f(w_\alpha))_{\alpha \in \Delta}$. By Exp. XXIII, 1.9, i is moreover a monomorphism.

Proposition 7.1.2. *If H is separated over S , $\mathcal{F}$ is representable and i is a closed immersion.*

The usual technique of relative representability⁽³²⁾ shows us that it suffices to prove that, given sections

$$v_\alpha, h_\alpha \in H(S), \quad \text{for } \alpha \in \Delta,$$

the S -schemes S' such that there exists an S' -homomorphism

$$f : G_{S'} \longrightarrow H_{S'}$$

⁽³⁰⁾**N.D.E.** Denoted by (étf) in Exp. IV 6.3. In other words, for every $s \in S$ there exist an open neighborhood U of s and a finite surjective étale morphism $V \longrightarrow U$ such that $B' \times_S V \simeq B \times_S V$.

⁽³¹⁾**N.D.E.** One may ask whether every smooth affine S -group, each of whose geometric fibers is a Borel subgroup of a semisimple group, is a Borel subgroup of a semisimple S -group. (We have removed the “counterexample” given in the original, for S = the scheme of dual numbers over a field k , which was erroneous, as M. Demazure pointed out to us.)

⁽³²⁾**N.D.E.** See for example the proof of XXII 5.8.1.

extending $(f_T)_{S'}$ and satisfying $f(u_\alpha) = v_\alpha$, $f(w_\alpha) = h_\alpha$, are precisely those which factor through a certain closed subscheme of S . One may evidently suppose S affine.

To simplify what follows, let us say that a morphism $Y \rightarrow X^{(33)}$ of affine schemes satisfies condition (L) if Y is the spectrum of an $\mathcal{O}(X)$ -algebra which is a free $\mathcal{O}(X)$ -module. It is clear that if one restricts to the category of affine schemes, a product or a composite of morphisms satisfying (L) satisfies (L), and that (L) is stable under extension of the base.

Lemma 7.1.3. *Suppose S affine, and let $\alpha \in \Delta$. Consider the morphism*

$$a : T \times_S T \rightarrow \mathbf{G}_{a,S}$$

defined set-theoretically by $a(t, t') = \alpha(t) + \alpha(t')$.

- (i) *a is faithfully flat and of finite presentation.*
- (ii) *Let R be the fibered square of a . The structure morphism $R \rightarrow S$ satisfies (L).*

PROOF. It is first clear that the morphism a is surjective, since $\alpha : T \rightarrow \mathbf{G}_{m,S}$ is so. It is trivial that a is of finite presentation. Since α satisfies (L), it suffices, to prove (i) and (ii), to show that the morphism

$$u : (\mathbf{G}_{m,S})^2 \rightarrow \mathbf{G}_{a,S}$$

defined set-theoretically by $u(x, y) = x + y$ satisfies (L). In other words, it suffices to verify that for every ring A , the ring $A[X, Y, X^{-1}, Y^{-1}]$ is a free module over its subring $A[X + Y]$.⁽³⁴⁾ Now $A[X, Y]$ is a free module with basis $\{1, X\}$ over the subring $B = A[X + Y, XY]$ (one has $X^2 - (X + Y)X + XY = 0$), hence $A[X^{\pm 1}, Y^{\pm 1}] = A[X, Y]_{XY}$ is free over $B_{XY} = A[X + Y, (XY)^{\pm 1}]$ with basis $\{1, X\}$, and free over $A[X + Y]$ with basis the elements $(XY)^p$ and $X(XY)^p$, for $p \in \mathbf{Z}$. $\square$

Lemma 7.1.4. *Let $\alpha \in \Delta$, and let $b : T \times_S T \rightarrow H$ be the morphism defined set-theoretically by*

$$b(t, t') = (\text{int}(f_T(t))v_\alpha)(\text{int}(f_T(t'))v_\alpha).$$

Let $f_\alpha : U_\alpha \rightarrow H$ be an S -morphism. The following conditions are equivalent.⁽³⁵⁾

- (i) *f_α is a group morphism; one has $f_\alpha(u_\alpha) = v_\alpha$ and*

$$f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(\alpha(t)x)$$

for all $t \in T(S')$, $x \in U_\alpha(S')$, $S' \rightarrow S$.

- (ii) *One has $f_\alpha(u_\alpha) = v_\alpha$ and the relation*

$$f_\alpha(a(t, t')u_\alpha) = b(t, t')$$

for all $t, t' \in T(S')$, $S' \rightarrow S$.

PROOF. If f_α satisfies (i), one has $f_\alpha(\alpha(t)u_\alpha) = \text{int}(f_T(t))v_\alpha$, which immediately implies (ii). Conversely, suppose (ii) holds and let us prove the various conditions of (i); let us first prove the last one. Since a is covering for (fpqc), it suffices to prove that if $t, t', t'' \in T(S')$, one has

$$f_T(t)f_\alpha(a(t', t'')u_\alpha)f_T(t)^{-1} = f_\alpha(\alpha(t)a(t', t'')u_\alpha);$$

now this can also be written

$$f_T(t)b(t', t'')f_T(t)^{-1} = b(tt', tt''),$$

⁽³³⁾**N.D.E.** We have corrected $X \rightarrow Y$ to $Y \rightarrow X$.

⁽³⁴⁾**N.D.E.** We have simplified the proof in the original.

⁽³⁵⁾**N.D.E.** In what follows, the group law of U_α is written additively, i.e. if one denotes by $p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$ the isomorphism such that $p_\alpha(1) = u_\alpha$ and if $x = p_\alpha(z)$, then $\alpha(t)x$ (resp. $a(t, t')u_\alpha$) denotes $p_\alpha(\alpha(t)z)$ (resp. $p_\alpha(a(t, t')) = \alpha(t)u_\alpha + \alpha(t')u_\alpha$).

a property evident from the definition of b . It remains to prove that f_α is a group morphism. Now the preceding property immediately gives

$$\begin{aligned} f_\alpha(\alpha(t)u_\alpha)f_\alpha(\alpha(t')u_\alpha) &= (f_T(t)f_\alpha(u_\alpha)f_T(t)^{-1}) \cdot (f_T(t')f_\alpha(u_\alpha)f_T(t')^{-1}) \\ &= b(t, t') = f_\alpha(\alpha(t)u_\alpha + \alpha(t)\alpha(t')u_\alpha). \end{aligned}$$

One therefore has $f_\alpha(x + x') = f_\alpha(x)f_\alpha(x')$ whenever x and x' are sections of the open subset $U_\alpha^\times$ of U_α , which is schematically dense; one then concludes by Exp. XVIII, 1.4. $\square$

7.1.5. — Let us provisionally fix an $\alpha \in \Delta$. The morphism a is faithfully flat and quasi-compact, hence $\mathbf{G}_{a,S}$ identifies with the quotient of $T \times_S T$ by the equivalence relation R defined by a . Let

$$R \rightrightarrows_{i_2}^{i_1} T \times_S T$$

be this equivalence relation.

For the morphism b to factor through a ,⁽³⁶⁾ it is necessary and sufficient that $b \circ i_1 = b \circ i_2$; in other words, if one denotes by K the kernel of the pair of morphisms

$$R \rightrightarrows_{b \circ i_2}^{b \circ i_1} H,$$

it is necessary and sufficient that $K = R$. Now H is assumed separated over S , hence K is a closed subscheme of R . Recall, on the other hand, that R is essentially free over S (7.1.3).

7.1.6. — According to the above, if S' is an S -scheme, for there to exist over S' an f_α satisfying the conditions of 7.1.4 (i) (and then necessarily unique), it is necessary and sufficient that $K_{S'} = R_{S'}$, and that the morphism f_α obtained satisfy $f_\alpha(u_\alpha) = v_\alpha$.

The first condition is equivalent to $S' \rightarrow S$ factoring through a certain closed subscheme of S (Exp. VIII, 6.4⁽³⁷⁾); if one replaces S by this closed subscheme, the second condition again defines a closed subscheme of S (equality of two sections of H , and H is assumed separated over S).

After replacing S by this closed subscheme, one may therefore suppose that there exists a morphism $f_\alpha : U_\alpha \rightarrow H$ satisfying the conditions of 7.1.4 (i). Taking the intersection of the subschemes of S obtained for each $\alpha \in \Delta$, one may suppose this condition satisfied for every $\alpha \in \Delta$.

7.1.7. — Likewise, consider for each $\alpha \in \Delta$ the two morphisms of S -groups

$$f_T \circ \text{int}(w_\alpha), \text{int}(h_\alpha) \circ f_T : T \rightarrow H.$$

Since H is separated over S and T essentially free over S , the same reasoning as before shows that, after replacing S by a closed subscheme, one may suppose that for every $\alpha \in \Delta$ one has

$$f_T \circ \text{int}(w_\alpha) = \text{int}(h_\alpha) \circ f_T.$$

7.1.8. — Using now the theorem on generators and relations (Exp. XXIII, 3.5), one sees that there exists a group homomorphism $f : G \rightarrow H$ satisfying the required conditions if and only if a certain finite set of algebraic relations between the sections $h_\alpha, v_\alpha, f_T(\alpha^*(-1))$ ($\alpha \in \Delta$) of H holds:

$$R_i((h_\alpha)_\alpha, (v_\alpha)_\alpha, (f_T(\alpha^*(-1)))_\alpha) = e, \quad i = 1, \dots, n.$$

Since H is separated over S , this again defines a closed subscheme of S , and the proof is complete.

⁽³⁶⁾**N.D.E.** I.e. for condition (ii) of 7.1.4 to hold.

⁽³⁷⁾**N.D.E.** See also the addition in Exp. VIB, 6.2.3.

Corollary 7.1.9. *Let S be a scheme, G a split reductive S -group, T its maximal torus, and H a separated S -group scheme. Let, on the other hand, $\mathcal{P}$ be a property of morphisms such that:*

- (i) *A closed immersion satisfies $\mathcal{P}$.*
- (ii) *The composite of two morphisms satisfying $\mathcal{P}$ also satisfies $\mathcal{P}$.*
- (iii) *$\mathcal{P}$ is stable under base change.*
- (iv) *The structure morphism $H \rightarrow S$ satisfies $\mathcal{P}$.*

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$$

is relatively representable by a separated morphism satisfying $\mathcal{P}$.⁽³⁸⁾

PROOF. Indeed, one may suppose G pinned; the structure morphism $H^{2n} \rightarrow S$ satisfies $\mathcal{P}$ and one concludes by 7.1.2. $\square$

Corollary 7.1.10. *Let S be a scheme, G a split reductive S -group, and H a smooth S -group scheme with affine fibers.⁽³⁹⁾ Then the functor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme locally of finite presentation and separated over S .*

PROOF. Indeed, since H is smooth over S , one can consider its identity component H^0 , which has affine fibers and is smooth, separated and of finite presentation over S (Exp. VIB, 3.10 and 5.5). Since G has connected fibers, one can replace H by H^0 . Since G and H are then of finite presentation, one may suppose S noetherian, and one applies 7.1.9 taking for $\mathcal{P}$ the property “of finite type”. But, by Exp. XV, 8.9, one knows that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$ is representable by an S -scheme separated and locally of finite type. $\square$

Remark 7.1.11. If H is affine over S , one can replace the reference to Exp. XV by Exp. XI, 4.2.

7.2. — General case

Proposition 7.2.1. *Let S be a scheme, G a reductive S -group, and T a maximal torus of G . Let, on the other hand, H be an S -group scheme such that the structure morphism $H \rightarrow S$ satisfies the following condition:*

(+) *Every point $s \in S$ has an open neighborhood U such that the morphism $H_U \rightarrow U$ is quasi-projective.*

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$$

is relatively representable by a morphism satisfying (+).

When G is splittable relative to T , one applies 7.1.9 taking for $\mathcal{P}$ property (+) above. When G is locally isotrivial, for example semisimple (4.2.2), one observes that the assertion of the theorem is local for the local finite étale topology (indeed, the property of being quasi-projective is local for the global finite étale topology and ensures effectiveness of descent for this topology, cf. SGA 1, VIII, 7.7). Finally, in the general case, one uses the following lemma:

Lemma 7.2.2. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , G' the derived group of G (Exp. XXII, 6.2), and $T' = T \cap G'$ the maximal torus of G' corresponding*

⁽³⁸⁾**N.D.E.** I.e. for every $S' \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$, with S' representable, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \times_S S'$ is representable and the morphism of S -schemes $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \times_S S' \rightarrow S'$ is separated and satisfies $\mathcal{P}$.

⁽³⁹⁾**N.D.E.** We have removed the hypothesis that H be quasi-separated, which is superfluous.

to T (Exp. XXII, 6.2.8). Then the diagram

$$\begin{array}{ccc} G & \longleftarrow & G' \\ \uparrow & & \uparrow \\ T & \longleftarrow & T' \end{array}$$

is an amalgamated sum in the category of S -sheaves of groups: i.e. for every S -sheaf of groups H , the following square is cartesian:

$$\begin{array}{ccc} \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) & \longrightarrow & \underline{\mathrm{Hom}}_{S\text{-gr.}}(G', H) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H) & \longrightarrow & \underline{\mathrm{Hom}}_{S\text{-gr.}}(T', H). \end{array}$$

PROOF. Indeed, if $\mathrm{rad}(G)$ is the radical of G , then $\mathrm{rad}(G) \subset T$, hence

$$\mathrm{rad}(G) \cap G' = \mathrm{rad}(G) \cap T' = K,$$

and multiplication in G induces isogenies (Exp. XXII, 6.2)

$$i : G' \times_S \mathrm{rad}(G) \longrightarrow G, \quad j : T' \times_S \mathrm{rad}(G) \longrightarrow T.$$

Let $f_{G'} : G' \longrightarrow H$, $f_T : T \longrightarrow H$, and $f_{T'} : T' \longrightarrow H$ be morphisms of S -groups such that $f_{G'}|_{T'} = f_T|_{T'} = f_{T'}$. Let us show that there exists a unique morphism of S -groups $f : G \longrightarrow H$ inducing $f_{G'}$ and f_T . Let $f_{\mathrm{rad}} = f_T|_{\mathrm{rad}(G)}$ ⁽⁴⁰⁾ and let

$$f_1 = f_{G'} \cdot f_{\mathrm{rad}} : G' \times_S \mathrm{rad}(G) \longrightarrow H.$$

For f to exist (and it will evidently be unique), it is necessary and sufficient that f_1 induce the identity on the kernel of i , i.e. that $f_{G'}$ and f_{rad} agree on K ; but the existence of f_T shows by the same reasoning that $f_{T'}$ and f_{rad} agree on K . It only remains to observe that $f_{G'}$ and $f_{T'}$ evidently agree on $K \subset T'$. $\square$

Reasoning now as in 7.1.10, one deduces from 7.2.1:

Corollary 7.2.3. *Let S be a scheme, G a reductive S -group, and H an S -group scheme, smooth and quasi-projective over S with affine fibers. Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme locally of finite presentation and separated over S .*

7.3. — Phenomena peculiar to characteristic 0 If G and H are two smooth S -groups, and $\mathfrak{g}$ and $\mathfrak{h}$ their Lie algebras, one has a canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h}),$$

where the right-hand side has an evident definition.

Proposition 7.3.1. *Let S be a scheme of characteristic 0 (i.e. a $\mathbf{Q}$ -scheme), G a reductive S -group, and H an S -group scheme smooth and quasi-projective over S with affine fibers.*

- (i) $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme smooth and separated over S .
- (ii) If G is semisimple, this scheme is affine and of finite presentation over S .
- (iii) If G is simply connected (Exp. XXII, 4.3.3), the canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$$

is an isomorphism.

Lemma 7.3.2. *Let k be a field of characteristic 0, G a reductive k -group, V a finite-dimensional k -vector space, and $G \longrightarrow \mathrm{GL}(V)$ a linear representation of G on V . One has*

$$H^0(G, V) = H^0(\mathfrak{g}, V), \quad H^i(G, V) = 0, \quad \text{for } i > 0.$$

⁽⁴⁰⁾**N.D.E.** We have replaced f_r by f_{rad} .

PROOF. The first equality is true in general for a smooth connected group;⁽⁴¹⁾ in the case of a reductive group, one can prove it as follows: one may suppose k algebraically closed, hence G splittable, hence G generated by subgroups isomorphic to $\mathbf{G}_{m,k}$,⁽⁴²⁾ and it suffices to verify the assertion for this group, which is easy.

From this first equality it follows that $H^0(G, V)$ is an exact functor in V when G is semisimple; $\mathfrak{g}$ is then indeed a semisimple Lie algebra and one applies [BLie], § I.6, Exercise 1 (b). The assertion remains true when G is reductive; indeed, introducing the radical R of G ⁽⁴³⁾ and the quotient G/R , which is semisimple, one has $H^0(G, V) = H^0(G/R, H^0(R, V))$, and $H^0(G, -)$ is the composite of two exact functors. Applying Exp. I, 5.3.1, one then deduces $H^i(G, V) = 0$ for $i > 0$. $\square$

Remark 7.3.3. Under the preceding conditions, if G is semisimple, one has $H^1(\mathfrak{g}, V) = H^2(\mathfrak{g}, V) = 0$, cf. [BLie], loc. cit. (b) and (d).

7.3.4. — Let us now prove the proposition. Already, by 7.2.3, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S -scheme locally of finite presentation and separated over S ; to show that it is smooth, it suffices to prove that it is infinitesimally smooth (Exp. XI, 1.8),⁽⁴⁴⁾ which follows from Exp. III, 2.8 (i) by 7.3.2. One has therefore proved (i).

Let us show that (ii) follows from (iii). It suffices first to prove that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is affine over S ; it will then be of finite presentation over S , since it is smooth over S by (i); in any case $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$ is representable by a closed subscheme of

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(\mathfrak{g}, \mathfrak{h}) \simeq \mathbf{W}(\mathfrak{g}^\vee \otimes_{\mathcal{O}_S} \mathfrak{h})$$

which is an affine S -scheme, and the desired conclusion appears when G is simply connected.

In the general case, one may suppose G split, hence $G \simeq \dot{\mathrm{E}}p_S(\mathcal{R})$; introducing the simply connected root datum $\mathrm{sc}(\mathcal{R})$ (Exp. XXI, 6.5.5), and using the existence theorem (Exp. XXV, 1.1), one constructs a simply connected S -group $\bar{G}$ and a central isogeny $\bar{G} \rightarrow G$. The kernel K of this isogeny is a finite diagonalizable S -group (hence a twisted constant S -group, S being of characteristic 0) and $\underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H)$ is (trivially) representable by a separated S -scheme (if $K \simeq (\mathbf{Z}/n\mathbf{Z})_S$, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H)$ is isomorphic to $\mathrm{Ker}(H \xrightarrow{n} H)$). One has an exact sequence of “pointed S -schemes”:

$$1 \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \xrightarrow{u} \underline{\mathrm{Hom}}_{S\text{-gr.}}(\bar{G}, H) \rightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H),$$

hence u is a closed immersion, hence $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is affine over S .

7.3.5. — Let us finally prove (iii). Reasoning as in the proof of (i) and using 7.3.3, one can show that the S -scheme $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$ is smooth over S . To prove (iii), one may therefore suppose that $S = \mathrm{Spec}(k)$, where k is an algebraically closed field of characteristic 0; it even suffices to prove that the morphism Lie is bijective on points with values in k and induces a bijection on the tangent spaces at two corresponding points. Let us first show that

$$\mathrm{Hom}_{k\text{-gr.}}(G, H) \rightarrow \mathrm{Hom}_{k\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$$

is bijective.

If $u : G \rightarrow H$ is a morphism of k -groups, the graph of u is a connected subgroup of $G \times_k H$ which is determined by its Lie algebra, which is the graph of $\mathrm{Lie}(u)$; the map is

⁽⁴¹⁾**N.D.E.** Indeed, it is clear that $V^G \subset V^\mathfrak{g}$. Moreover, $H = \mathrm{Centr}_G(V^\mathfrak{g})$ is a closed subgroup of G , smooth since $\mathrm{char}(k) = 0$; by Exp. II 5.3.1, one has $\mathrm{Lie}(H) = \mathrm{Centr}_\mathfrak{g}(V^\mathfrak{g}) = \mathfrak{g}$, and since H is smooth and G connected this implies $H = G$, whence $V^\mathfrak{g} \subset V^G$ (see also [DG70], § II.6, Prop. 2.1 (c)).

⁽⁴²⁾**N.D.E.** Indeed, the union of the maximal tori of G is dense in G , cf. Bible, § 6.5, Th. 5 (= [Ch05], § 6.6, Th. 6).

⁽⁴³⁾**N.D.E.** Note that R is a torus.

⁽⁴⁴⁾**N.D.E.** I.e. for every local Artinian S -scheme S' , with closed point s' , every morphism of $\kappa(s')$ -groups $G_{s'} \rightarrow H_{s'}$ lifts to a morphism of S' -groups $G_{S'} \rightarrow H_{S'}$. By Exp. III 2.8, this follows from the vanishing of $H^2(G_{s'}, V)$, where $V = \mathrm{Lie}(H_{s'}/\kappa(s'))$.

therefore injective. Conversely, if $v : \mathfrak{g} \rightarrow \mathfrak{h}$ is a Lie algebra morphism, the graph $\mathfrak{k}$ of v is a Lie subalgebra of $\mathfrak{g} \times \mathfrak{h}$, isomorphic to $\mathfrak{g}$. In particular, since $\mathfrak{g}$ is its own derived algebra, the same is true of $\mathfrak{k}$ and therefore, by a theorem of Chevalley ([Ch51], § 14, Th. 15), $\mathfrak{k}$ is the Lie algebra of a connected subgroup K of $G \times_k H$.⁽⁴⁵⁾ Moreover, since $\mathfrak{k} \simeq \mathfrak{g}$ is semisimple, G is a semisimple k -group. Since

$$\dim(K) = \dim(\mathfrak{k}) = \dim(\mathfrak{g}) = \dim(G)$$

and since $K \cap H$ is finite (because its Lie algebra is zero), the canonical morphism $\mathrm{pr}_1 : K \rightarrow G$ is finite and dominant; since G is connected, pr_1 is surjective; it is therefore an isogeny. Since G is simply connected and K semisimple, by Exp. XXI 6.2.7, pr_1 is an isomorphism, hence K is the graph of a morphism of k -groups $u : G \rightarrow H$ such that $\mathrm{Lie}(u) = v$.

Finally, let $u : G \rightarrow H$ be a morphism of k -groups. The tangent space to $\underline{\mathrm{Hom}}_{k\text{-gr.}}(G, H)$ at u identifies with $Z^1(G, \mathfrak{h})$ (cf. Exp. II, 4.2; G acts on $\mathfrak{h}$ by $\mathrm{Ad}_H \circ u$); in the same way, one can prove that the tangent space to $\underline{\mathrm{Hom}}_{k\text{-Lie alg.}}(\mathfrak{g}, \mathfrak{h})$ at $\mathrm{Lie}(u)$ identifies with $Z^1(\mathfrak{g}, \mathfrak{h})$. We must therefore prove that the canonical map $Z^1(G, \mathfrak{h}) \rightarrow Z^1(\mathfrak{g}, \mathfrak{h})$ is bijective. Now one has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{h}^G & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(G, \mathfrak{h}) \longrightarrow H^1(G, \mathfrak{h}) \\ & & \downarrow & & \parallel & & \downarrow \\ 0 & \longrightarrow & \mathfrak{h}^{\mathfrak{g}} & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(\mathfrak{g}, \mathfrak{h}) \longrightarrow H^1(\mathfrak{g}, \mathfrak{h}) \end{array}$$

and by 7.3.2 and 7.3.3 one has $\mathfrak{h}^G = \mathfrak{h}^{\mathfrak{g}}$ and $H^1(G, \mathfrak{h}) = 0 = H^1(\mathfrak{g}, \mathfrak{h})$, whence the desired conclusion.

Remark 7.3.6. It is likely that if S is a locally noetherian scheme, G a simply connected semisimple S -group, H a smooth S -group scheme, and $\widehat{G}$ and $\widehat{H}$ their formal completions along the identity section, every homomorphism $\widehat{G} \rightarrow \widehat{H}$ comes from a unique homomorphism from G to H , which would generalize 7.3.1 (iii). When S is the spectrum of a field k and H is affine and of finite type, this follows from a theorem of Dieudonné ([Di57], § 15, Th. 4).⁽⁴⁶⁾

7.4. — An example As an example, we shall determine

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S}).$$

7.4.1. — Recall (Exp. XX, no. 5) that $\mathrm{SL}_{2,S}$ is the S -group scheme consisting of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ over S satisfying $ad - bc = 1$. A maximal torus T is defined by the monomorphism $\alpha^* : \mathbf{G}_{m,S} \rightarrow \mathrm{SL}_{2,S}$

$$\alpha^*(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}.$$

⁽⁴⁵⁾**N.D.E.** See also [Bo91], II 7.9. Moreover, we have added the following sentence.

⁽⁴⁶⁾**N.D.E.** See Exp. VIIB for the definition of the formal groups $\widehat{G}$ and $\widehat{H}$. Suppose that $S = \mathrm{Spec}(k)$, where k is a field. By loc. cit., 2.2.1, giving a morphism of formal k -groups $v : \widehat{G} \rightarrow \widehat{H}$ is equivalent to giving a morphism of Hopf k -algebras $\phi : \mathrm{Dist}(G) \rightarrow \mathrm{Dist}(H)$, where $\mathrm{Dist}(G)$ denotes the algebra of distributions (at the origin) of G , cf. Exp. VIIA, 2.1 or [DG70], § II.4, 6.1 or [Ja87], I 7.7 (it is called the “hyperalgebra” of G in [Di57] and [Ta75]). Theorem 4 of [Di57] (see also [Ta75], 0.3.4 (f) and (g)) generalizes in this context the theorem of Chevalley used in 7.3.5; one thus obtains that there exists a connected closed k -subgroup K of $G \times H$ such that $\widehat{K}$ equals the graph of v ; since $\mathrm{Dist}(K) \rightarrow \mathrm{Dist}(G)$ is an isomorphism, $K \rightarrow G$ is a finite étale morphism. One then deduces that K is semisimple and then, by Exp. XXI 6.2.7, that $K \rightarrow G$ is an isomorphism (since G is simply connected); see also [Ta75], 1.8 and 2.2. More generally, loc. cit. studies the k -groups G with property (SC): every finite étale morphism of k -groups $G' \rightarrow G$, with G' connected, is an isomorphism. Finally, observe that the above shows that every finite-dimensional $\mathrm{Dist}(G)$ -module V is, uniquely, a G -module; for an extension to the case of a split reductive k -group G (or even of a Borel subgroup of G), see [Ja87], II 1.20 (and the references there).

A root of G relative to T is defined by $\alpha(\alpha^*(z)) = z^2$, a corresponding monomorphism

$$p : \mathbf{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}$$

being defined by

$$p(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Finally, the representative of the Weyl group corresponding to $u = p(1)$ is

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Recall moreover (Exp. XX, 6.2) that $\mathrm{SL}_{2,S}$ is generated by T , $p(\mathbf{G}_{a,S})$ and w , subject to the relations

$$\begin{aligned} \alpha^*(z)p(x)\alpha^*(z^{-1}) &= p(z^2x), \\ w\alpha^*(z)w^{-1} &= \alpha^*(z^{-1}), \\ w^2 &= \alpha^*(-1), \\ (wu)^3 &= 1. \end{aligned}$$

7.4.2. — Let f be an endomorphism of $\mathrm{SL}_{2,S}$. It first defines a homomorphism $f \circ \alpha^* : \mathbf{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$. The kernel $\mathrm{Ker}(f \circ \alpha^*)$ is a closed subgroup of $\mathbf{G}_{m,S}$, hence is locally on S equal to a $\mu_{n,S}$ ($n \geq 1$) or to $\mathbf{G}_{m,S}$. After restricting S , one may therefore suppose that there exist an $n \in \mathbf{N}$ and a monomorphism

$$f' : \mathbf{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$$

such that $f \circ \alpha^*(z) = f'(z^n)$.

By conjugacy of the monomorphisms $\mathbf{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$, one can, after an extension of the base covering for the étale topology, find a section g of G such that $f \circ \alpha^*(z) = \mathrm{int}(g) \circ \alpha^*(z^n)$. Transforming f by $\mathrm{int}(g)$, one is therefore reduced to the case where there exists an $n \in \mathbf{N}$ such that $f \circ \alpha^*(z) = \alpha^*(z^n)$.

7.4.3. — Consider now the morphism

$$f \circ p : \mathbf{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}.$$

It satisfies the condition

$$\alpha^*(z^n)(f \circ p)(x)\alpha^*(z^n)^{-1} = (f \circ p)(z^2x).$$

If $n = 0$, it follows at once that $f \circ p$ is invariant under homotheties of $\mathbf{G}_{a,S}$, hence constant. If $n \neq 0$, one can apply Exp. XXII, 4.1.9; $x \longmapsto x^n$ is an endomorphism of $\mathbf{G}_{a,S}$, and there exists a $\lambda \in \mathbf{G}_{m,S}$ such that

$$f \circ p(x) = p(\lambda x^n);$$

this relation is moreover valid for $n = 0$, taking $\lambda = 1$. After again extending S , one can find a $z \in \mathbf{G}_{m,S}$ such that $z^2 = \lambda$. Replacing f by $\mathrm{int}(\alpha^*(z)) \circ f$, one is therefore reduced to the case where one has

$$f \circ \alpha^*(z) = \alpha^*(z^n), \quad f \circ p(x) = p(x^n).$$

7.4.4. — Finally, one must have $f(w)\alpha^*(z^n)f(w)^{-1} = \alpha^*(z^n)^{-1}$ and $(f(w)u)^3 = 1$. By Exp. XX, 3.8, this implies $f(w) = w^n$. Since G is generated by T , $p(\mathbf{G}_{a,S})$ and w , this implies that for every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(S')$, $S' \longrightarrow S$, one has

$$f \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

7.4.5. — Summarizing the preceding discussion, one sees that locally on S for the étale topology, one can find for every endomorphism f of $\mathrm{SL}_{2,S}$ an (inner) automorphism ϕ of

$\mathrm{SL}_{2,S}$, and an integer $n \geq 0$ such that $f = \phi \circ F_n$, where

$$F_n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

Observe that if $f = \phi \circ F_n$, the integer n is determined by a fiber f_s of f , for example by $\mathrm{Ker}(f_s)$. It follows that n is a locally constant function on S .

7.4.6. — One deduces immediately that if f is an endomorphism of $\mathrm{SL}_{2,S}$, then S decomposes canonically as a sum of open and closed subschemes S_0, S_1, S_{p^n} (where p^n ranges over the powers > 0 of prime numbers) such that:

- (i) f_{S_0} is the zero morphism,
- (ii) f_{S_1} is an isomorphism (= an inner automorphism),
- (iii) S_{p^n} is of characteristic p and $f_{S_{p^n}}$ decomposes uniquely in the form $\phi \circ F_p^n$, where ϕ is an inner automorphism and F_p the Frobenius endomorphism of $\mathrm{SL}_{2,\mathbf{F}_p}$.

7.4.7. — In other words, $\underline{\mathrm{Hom}}_{\mathbf{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbf{Z}}, \mathrm{SL}_{2,\mathbf{Z}})$ is the sum scheme:

- (i) of a scheme isomorphic to $\mathrm{Spec}(\mathbf{Z})$,
- (ii) of a scheme isomorphic to $\underline{\mathrm{Aut}}_{\mathbf{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbf{Z}}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbf{Z}})$,
- (iii) for each prime number p and each integer $n > 0$, of a scheme isomorphic to $\underline{\mathrm{Aut}}_{\mathbf{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbf{F}_p}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbf{F}_p})$.

7.4.8. — It follows in particular that

- (i) $\underline{\mathrm{Hom}}_{\mathbf{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbf{F}_p}, \mathrm{SL}_{2,\mathbf{F}_p})$ has infinitely many connected components, and therefore *is not quasi-compact*.
- (ii) If S is a scheme of mixed characteristics, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S})$ *is not flat* over S .

8. Appendix: Cohomology of a smooth group over a henselian ring. Cohomology and the functor $\mathbb{I}$

Proposition 8.1. *Let S be a henselian local scheme, s its closed point, and G a smooth S -group scheme such that every finite subset of G is contained in an affine open subset.* Then*

- (i) *If P is a principal homogeneous bundle under G , there exists a finite surjective étale morphism $S' \rightarrow S$ trivializing P . One has $\mathrm{Fib}(S, G) \simeq H_{\mathrm{\acute{e}t}}^1(S, G)$.*
- (ii) *For every finite surjective étale morphism $S' \rightarrow S$, the canonical map*

$$H^1(S'/S, G) \rightarrow H^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is bijective.

- (iii) *The canonical map*

$$\mathrm{Fib}(S, G) \rightarrow \mathrm{Fib}(\kappa(s), G_s)$$

is bijective.

8.1.2. — If K is a finite separable extension of $\kappa(s)$, there exists a finite surjective étale morphism $S' \rightarrow S$ such that $K \simeq S' \otimes_S \kappa(s)$.⁽⁴⁷⁾ If P is a principal homogeneous bundle under G , then P is smooth over S , hence P_s smooth over $\kappa(s)$; there therefore exists a finite separable extension K of $\kappa(s)$ such that P_K has a section (cf. EGA IV₄, 17.15.10). Representing K as indicated above, one sees that $P_{S'}$ has a section by “Hensel’s lemma” (cf. EGA IV₄, 18.5.17), which proves the first part of (i).

Conversely, if P is a principal homogeneous *sheaf* under G for the étale topology, there exists a finite surjective étale morphism $S' \rightarrow S$ trivializing P (indeed every covering family

*This last condition is in fact unnecessary (cf. A. Grothendieck, *Groupe de Brauer III*, in *Dix Exposés sur la cohomologie des schémas*, North-Holland, 1968, Theorem 11.7 and Remarks 11.8.3).

⁽⁴⁷⁾**N.D.E.** Note that S' is still local and henselian. Moreover, we have preserved the numbering of the original: there is no no. 8.1.1.

of a henselian local scheme for the étale topology is refined by a family consisting of a single morphism $S' \rightarrow S$ of the required form).

By the descent hypothesis made on G , P is representable (SGA 1, VIII, 7.6), which completes the proof of (i), and shows that (ii) implies (iii). It therefore only remains for us to prove (ii).

8.1.3. — The map of (ii) is injective: let P and Q be two principal homogeneous bundles under G trivialized by S' . Consider the S -sheaf of groups $H = \underline{\text{Isom}}_{G\text{-bundles}}(P, Q)$; since $H_{S'}$ is isomorphic to $G_{S'}$, H is representable, by the second hypothesis on G , cf. above. If $H(\kappa(s)) \neq \emptyset$, then $H(S) \neq \emptyset$ by Hensel's lemma, hence P and Q are isomorphic.

8.1.4. — Let us finally prove that the map of (ii) is surjective, or again that the canonical map

$$Z^1(S'/S, G) \rightarrow Z^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is surjective. Let $\mathbf{Z}^1$ be the S -functor defined by

$$\mathbf{Z}^1(T) = Z^1(S' \times_S T/T, G_T);$$

the preceding map identifies with the map

$$\mathbf{Z}^1(S) \rightarrow \mathbf{Z}^1(\kappa(s));$$

by another application of Hensel's lemma, it suffices to prove that $\mathbf{Z}^1$ is representable by a smooth S -scheme.

8.1.5. — Let us prove that $\mathbf{Z}^1$ is representable by an S -scheme locally of finite presentation. Let $\mathbf{C}^i$, $i = 0, 1, \dots$, be the S -functor defined by

$$\mathbf{C}^i(T) = C^i(S' \times_S T/T, G_T),$$

i.e.

$$\mathbf{C}^i(T) = G((S' \times_S T/T)^{i+1}) = G((S'/S)^{i+1} \times_S T),$$

or again

$$\mathbf{C}^i = \underline{\text{Hom}}_S((S'/S)^{i+1}, G).$$

Since $\mathbf{Z}^1$ is obtained from $\mathbf{C}^1$ and $\mathbf{C}^2$ by fibered products, it suffices to prove that $\mathbf{C}^i$, $i = 1, 2$, is representable by an S -scheme locally of finite presentation.

8.1.6. — If $T \rightarrow S$ is a finite surjective étale morphism splitting S' , then

$$\mathbf{C}^i \times_S T = \underline{\text{Hom}}_T((S' \times_S T/T)^{i+1}, G_T)$$

is representable by a product of n copies of G_T , where n is the degree of $(S'/S)^{i+1}$. Applying the hypothesis on G once more, one deduces that $\mathbf{C}^i$ is indeed representable by an S -scheme locally of finite presentation (SGA 1, VIII, loc. cit.).

8.1.7. — To prove that $\mathbf{Z}^1$ is smooth, we must now, by definition, prove that if T is an *affine* S -scheme, and T_0 the closed subscheme defined by a square-zero ideal $\mathcal{J}$, the canonical map

$$\mathbf{Z}^1(T) \rightarrow \mathbf{Z}^1(T_0)$$

is surjective. Since G is smooth, the canonical map $G(T) \rightarrow G(T_0)$ is surjective, and it suffices to prove that the canonical map

$$H^1(S' \times_S T/T, G_T) \rightarrow H^1(S' \times_S T_0/T_0, G_{T_0})$$

is bijective.

Changing the notation slightly and generalizing the hypotheses, it now suffices for us to prove:

Lemma 8.1.8. *Let S and S' be two affine schemes, $S' \rightarrow S$ a faithfully flat morphism, $\mathcal{J}$ a square-zero ideal on S , S_0 the closed subscheme it defines, and G a smooth S -group. Put $S'_0 = S' \times_S S_0$, $G_0 = G \times_S S_0$. The canonical map*

$$H^1(S'/S, G) \rightarrow H^1(S'_0/S_0, G_0)$$

is bijective.

Remark 8.1.9. If one assumes G commutative, the same assertion holds for all the H^i , $i > 0$, with an analogous proof.

PROOF. Let $\mathcal{M}_0$ be the quasi-coherent $\mathcal{O}_{S_0}$ -module $\mathcal{M}_0 = \text{Lie}(G_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$. For each S_0 -prescheme T_0 , put $M_0(T_0) = H^0(T_0, \mathcal{M}_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0})$, and let

$$M = \prod_{S_0/S} M_0 \quad \text{and} \quad \bar{G} = \prod_{S_0/S} G_0$$

be the S -functors of groups defined by $M(T) = M_0(T_0)$ and $\bar{G}(T) = G_0(T_0)$, where $T_0 = T \times_S S_0$. By Exp. III, 0.9 and (0.6.2), there exists for every affine S -scheme T an exact sequence, functorial in T :

$$1 \rightarrow M(T) \rightarrow G(T) \rightarrow \bar{G}(T) \rightarrow 1.$$

We have to study the map

$$H^1(S'/S, G) \rightarrow H^1(S'_0/S_0, G_0) = H^1(S'/S, \bar{G}).$$

Suppose first that G is commutative. One then has an exact cohomology sequence

$$H^1(S'/S, M) \rightarrow H^1(S'/S, G) \rightarrow H^1(S'/S, \bar{G}) \rightarrow H^2(S'/S, M);$$

but

$$H^i(S'/S, M) = H^i(S'_0/S_0, M_0) = H^i(S'_0/S_0, \mathcal{M}_0),$$

and one knows (TDTE I, B, Lemma 1.1) that $H^i(S'_0/S_0, \mathcal{M}_0) = 0$ for $i \neq 0$.

If now G is not commutative, we must use the nonabelian cohomology exact sequence. If $u \in Z^1(S'/S, G)$, one knows that the elements of $H^1(S'/S, G)$ having the same image in $H^1(S'/S, \bar{G})$ as the class of u are in the image of the corresponding coboundary map:

$$H^1(S'/S, M_u) \rightarrow H^1(S'/S, G),$$

where M_u is the S -functor M “twisted by u ”. Likewise, if v is an element of $Z^1(S'/S, \bar{G})$, there exists a “coboundary”

$$\Delta(v) \in H^2(S'/S, M_v),$$

where M_v is the S -functor M “twisted by v ”, such that $\Delta(v) = 0$ if and only if the class of v is in the image of $H^1(S'/S, G)$. It suffices for us to prove that $H^1(S'/S, M_u) = H^2(S'/S, M_v) = 0$.

Now recall (Exp. III 0.8) that the action of $\bar{G}$ on M defined by the exact sequence is nothing other than the one deduced functorially from the adjoint representation of G_0 :

$$\text{ad} : G_0 \rightarrow \underline{\text{Aut}}_{\mathcal{O}_{S_0}}(\text{Lie}(G_0/S_0)).$$

The element u (resp. v) therefore acts on $M_{S' \times_S S'}$ through an $S'_0 \times_{S_0} S'_0$ -automorphism of $\text{Lie}(G_0/S_0)$. Since u (resp. v) is a cocycle, this automorphism is a descent datum; denote by $\mathcal{L}_u$ (resp. $\mathcal{L}_v$) the quasi-coherent $\mathcal{O}_{S_0}$ -module obtained. One verifies at once that for $T \rightarrow S$ one has

$$M_u(T) = H^0(T_0, \mathcal{L}_u \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0})$$

and the same relation on replacing u by v . One therefore has

$$H^1(S'/S, M_u) = H^1(S'_0/S_0, \mathcal{L}_u \otimes \mathcal{J}),$$

$$H^2(S'/S, M_v) = H^2(S'_0/S_0, \mathcal{L}_v \otimes \mathcal{J}),$$

and both do indeed vanish by the result already used. $\square$

Proposition 8.2. *Let $\mathcal{C}$ be a category possessing fibered products, equipped with a topology coarser than the canonical topology, $S' \rightarrow S$ a morphism of $\mathcal{C}$, G' an S' -sheaf of groups, and G the S -sheaf of groups $\prod_{S'/S} G'$. Let $H_S^1(S', G') \subset H^1(S', G')$ be the set of classes of principal homogeneous sheaves under G' which are trivialized by a sieve of S' obtained by base change from a suitable covering sieve of S . The canonical map $H^1(S, G) \rightarrow H^1(S', G')$ defined by the functor*

$$P \mapsto P \times_S S'$$

induces a bijection

$$H^1(S, G) \xrightarrow{\sim} H_S^1(S', G');$$

the inverse bijection is defined by the functor $P' \mapsto \prod_{S'/S} P'$.

PROOF. For every object X of $\mathcal{C}/S$, one has by definition an isomorphism functorial in X

$$\mathrm{Hom}_S(X, G) \xrightarrow{\sim} \mathrm{Hom}_{S'}(X \times_S S', G').$$

One therefore has for each S -object T a bijection functorial in T

$$H^1(T/S, G) \xrightarrow{\sim} H^1(T'/S', G').$$

Replacing now the single morphism $T \rightarrow S$ by an arbitrary covering family of S and passing to the inductive limit, one deduces the first part of the statement. The second part follows without difficulty. $\square$

Lemma 8.3. *Under the conditions of 8.2, the assertion $H_S^1(S', G') = H^1(S', G')$ is local on S : suppose that there exists a covering family $\{S_i \rightarrow S\}$ such that for every i one has $H_{S_i}^1(S' \times_S S_i, G') = H^1(S' \times_S S_i, G')$. Then $H_S^1(S', G') = H^1(S', G')$.*

PROOF. Indeed, let P' be a principal homogeneous sheaf under G' . Put

$$P'_i = P' \times_{S'} (S' \times_S S_i);$$

by the hypothesis, there exists a covering family $\{S_{ij} \rightarrow S_i\}$ such that for each j , $P' \times_{S'} (S' \times_S S_{ij})$ has a section. But $\{S_{ij} \rightarrow S\}$ is a covering family of S , and P' is indeed trivialized by the covering family of S' obtained from it by base change. $\square$

Proposition 8.4. *Let $S' \rightarrow S$ be a finite étale morphism of schemes. Let G' be an S' -sheaf of groups, and G the S -sheaf of groups $\prod_{S'/S} G'$. For the étale (resp. local finite étale, resp. (fpqc)) topology, the functors*

$$P \mapsto P \times_S S' \\ \prod_{S'/S} P' \longleftarrow P'$$

induce mutually inverse bijections:

$$H^1(S, G) \simeq H^1(S', G').$$

PROOF. By 8.2, it suffices to show that $H_S^1(S', G') = H^1(S', G')$. By 8.3, it suffices to do so locally for the local finite étale topology; one may therefore suppose that S' is a finite direct sum of copies of S , say I_S , where I is a suitable finite set. Then G' is given by a family $(G_i)_{i \in I}$ of sheaves on S , and

$$H^1(S', G') \simeq \prod_{i \in I} H^1(S, G_i).$$

On the other hand

$$H^1(S, G) \simeq \prod_{i \in I} H^1(S, G_i),$$

whence, by 8.2, $H_S^1(S', G') = H^1(S', G')$. $\square$

Remarks 8.5. One can interpret 8.2 and 8.3 by means of the following exact sequence (f is the given morphism $S' \rightarrow S$)

$$1 \rightarrow H^1(S, f_*(G')) \rightarrow H^1(S', G') \rightarrow H^0(S, R^1 f_*(G')).$$

In the commutative case, this exact sequence follows from the Leray spectral sequence; it is still valid in the noncommutative case (cf. Giraud's thesis⁽⁴⁸⁾).

In this form, one sees that the result is still valid if f is merely assumed *finite*, or simply *integral*, the topology being the étale topology, since for every G' one has

$$R^1 f_*(G') = \text{final sheaf},$$

by SGA 4, VIII, 5.3.⁽⁴⁹⁾

On the other hand, this result becomes false if one takes a topology such as (fpqc) or (fppf), even if $S = \text{Spec}(k)$, with k an algebraically closed field of characteristic $p \neq 0$, $S' = \text{Spec}(k[t]/t^2)$, and $G' = \mu_p$ or α_p .

Likewise, 8.2 becomes false, even for the étale topology, if one removes the hypothesis that f is finite, as one sees by taking for f an open immersion; for example, if $S = \text{Spec}(V)$, with V a complete discrete valuation ring with algebraically closed residue field, S' being the open subset induced at the generic point, and G' the constant group $(\mathbf{Z}/n\mathbf{Z})_{S'}$, with n prime to the residue characteristic of V , one has $H^1(S, G) = 0$, $H^1(S', G') \neq 0$. Replacing moreover S' by $S \amalg S'$, one deduces an analogous example, with $S' \rightarrow S$ étale and surjective, hence covering for the topology in question.

⁽⁴⁸⁾**N.D.E.** See paragraph V 3.1.4 of [Gi71].

⁽⁴⁹⁾**N.D.E.** This reference also points to paragraph V 3.1.4 of [Gi71].

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⁽⁵⁰⁾**N.D.E.** To these three references, appearing in the original, we have added the following references.